NETWORK / SOC ENGINEER L1 · INTERVIEW & FIELD GUIDE · PART 1 OF 2

Switching, Routing, TCP/IP & Firewall Troubleshooting

Theory · Configuration · Packet flow · Wireshark captures explained · Scenario questions · Customer-network troubleshooting

OSI & TCP/IP | ARP / ICMP | TCP / UDP | DNS / DHCP | Subnetting / NAT | VLAN / STP / EtherChannel | OSPF / BGP / HSRP | Palo Alto / IPSec / GlobalProtect | Wireshark

ABOUT THE WIRESHARK SCREENSHOTS

The capture images in this guide are Wireshark-style renderings of realistic example packets (field names, values and colours follow what Wireshark shows). They are for learning to read captures. They are not live screenshots from a real network, so IPs and MACs are lab examples. Open the same traffic in your own lab (GNS3 / EVE-NG / Packet Tracer + Wireshark) and compare.

Contents: Part 1 of 2

1	OSI, TCP/IP and end-to-end packet flow
2	Ethernet, MAC and ARP
3	IP, ICMP, ping and traceroute
4	TCP and UDP
5	DNS and DHCP
6	Subnetting and NAT/PAT
7	Switching: VLAN, trunk, STP, EtherChannel, switch security
8	Routing: static, OSPF, BGP, HSRP

Part 2 contains: 9. Firewall: Palo Alto flow, NAT, IPSec, GlobalProtect, Panorama; 10. Wireshark field guide and filters; 11. How routers communicate: hop-by-hop worked example; 12. Troubleshooting a customer network (architecture + scenarios); 13. Rapid-fire interview Q&A; 14. Last-day revision plan and cheat sheet.

HOW TO USE THIS GUIDE

Order of study: chapters 1, 4, 2, 7 (OSI, TCP, ARP, switching) first, as you asked, then 8, 5, 3, 6, then Part 2 (firewall, Wireshark, router communication, customer troubleshooting, Q&A). Every chapter ends with scenario questions so you practise answering aloud.

1. OSI, TCP/IP and end-to-end packet flow

The model you use to localise every fault

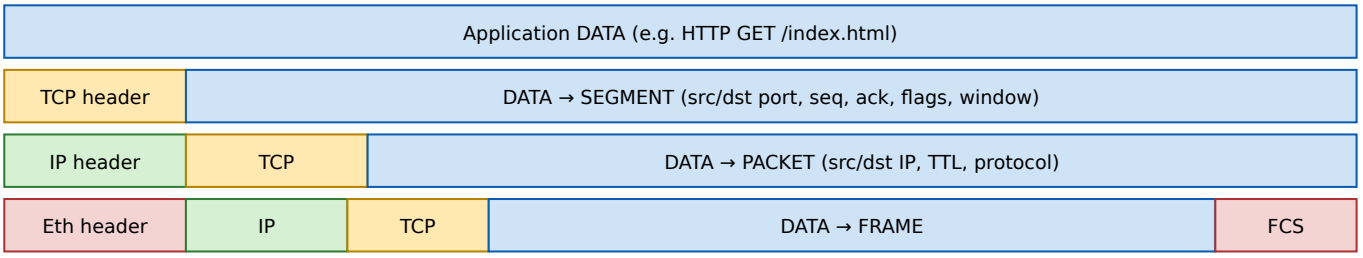
1.1 Theory: OSI vs TCP/IP

#	OSI layer	PDU	Does what	Examples / devices
7	Application	Data	Interface for apps, user-facing protocols	HTTP, HTTPS, DNS, DHCP, SSH, SMTP · L7 firewall (App-ID)
6	Presentation	Data	Encoding, encryption, compression	TLS/SSL, JPEG, ASCII
5	Session	Data	Set up, maintain, tear down sessions	NetBIOS, RPC, TLS session
4	Transport	Segment (TCP) / Datagram (UDP)	End-to-end delivery, ports, reliability, flow control	TCP, UDP · stateful firewall, load balancer
3	Network	Packet	Logical addressing, routing	IPv4/IPv6, ICMP, OSPF, BGP · router, L3 switch
2	Data Link	Frame	MAC addressing, switching, error detection (FCS), VLAN	Ethernet, 802.1Q, STP, ARP* · switch, bridge, NIC
1	Physical	Bits	Signals on the medium	Cable, fibre, SFP, hub, repeater

TCP/IP model has 4 layers: **Application** (OSI 5-7), **Transport** (4), **Internet** (3), **Network Access / Link** (1-2). Mnemonic top to bottom: *All People Seem To Need Data Processing*.

> **TIP**
> *ARP sits between L2 and L3. It resolves an L3 address (IP) to an L2 address (MAC). Interviewers like this one: say "it is usually placed at layer 2/3" and explain what it does.

1.2 Encapsulation and decapsulation



Frame header holds src/dst MAC + EtherType (0x0800 IPv4, 0x0806 ARP, 0x8100 802.1Q). FCS lets the receiver detect corruption (CRC errors).

Each layer wraps the data from the layer above. The receiver strips headers in reverse order.

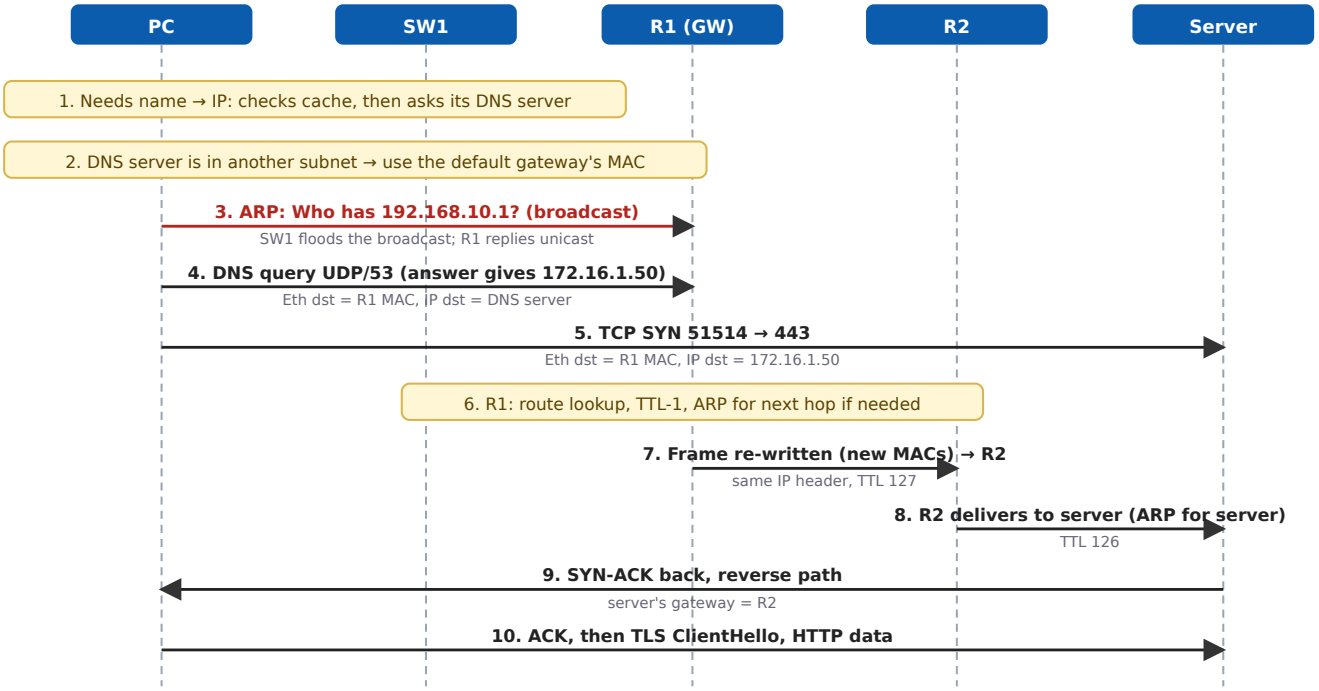
1.3 Key rule: what changes at each hop

- **MAC addresses change at every router hop** (L2 is local to the segment).
- **IP addresses stay the same end-to-end** unless NAT rewrites them (firewall/edge router).
- **TTL decreases by 1 at every router.** At 0 the router drops the packet and sends ICMP Time Exceeded (type 11).
- **Ports stay the same** unless PAT rewrites the source port.

1.4 Complete packet flow: PC to web server in another network

Scenario: PC `192.168.10.10/24` opens `https://app.corp.local` hosted at `172.16.1.50`. Default gateway is R1.

End-to-end flow (switch SW1 forwards frames by MAC table, without changing them)



1.5 Troubleshooting with the OSI model

Layer	Check	Tool / command	Typical finding
1 Physical	Link light, cable, SFP, speed/duplex	`show interfaces status`, `show interfaces g0/1`	down/down, CRC errors, input errors, duplex mismatch
2 Data link	VLAN, trunk allowed list, MAC table, STP state, port-channel	`show vlan brief`, `show mac address-table`, `show spanning-tree`	Wrong VLAN, port blocked by STP, err-disabled
3 Network	IP, mask, gateway, route, ACL, NAT	`ipconfig`, `show ip route`, `ping`, `traceroute`	No route, wrong mask, asymmetric route
4 Transport	Port open? Handshake completing? Firewall policy?	`telnet host 443`, `nc -zv`, Wireshark	SYN with no SYN-ACK, RST, retransmits
5-7 Application	DNS, certificate, app response, auth	`nslookup`, `curl -v`, browser dev tools	NXDOMAIN, TLS alert, HTTP 5xx

INTERVIEW ANGLE
"User cannot access an application. How do you troubleshoot?" Answer in layers: confirm scope (one user or many), check link and IP config, ping the gateway, ping the destination by IP, test DNS by name, test the port (TCP handshake), check firewall session and policy, capture packets if needed. Then say how you document and escalate. Interviewers want a structured method, not random commands.

1.6 Scenario questions

Question	Strong answer outline
Which layer does a switch, router, firewall work at?	Switch L2 (L3 if multilayer), router L3, traditional firewall L3/L4, next-gen firewall up to L7 (App-ID, URL, content).
Difference between a frame, packet and segment?	L2, L3 and L4 PDUs. Frame has MACs and FCS, packet has IPs and TTL, segment has ports and sequence numbers.
Ping works but the website does not. What next?	L3 is fine. Check DNS (nslookup), then TCP port reachability, firewall policy/session, TLS/certificate, proxy, application.
Can two hosts in different VLANs talk without a router?	No. Different VLAN means different broadcast domain/subnet, so you need an L3 device (router-on-a-stick or SVI).
Why do MAC addresses change per hop but IPs do not?	MAC is only meaningful on the local segment; the router strips the L2 header and builds a new one for the next segment. IP identifies the end hosts.

2. Ethernet, MAC and ARP

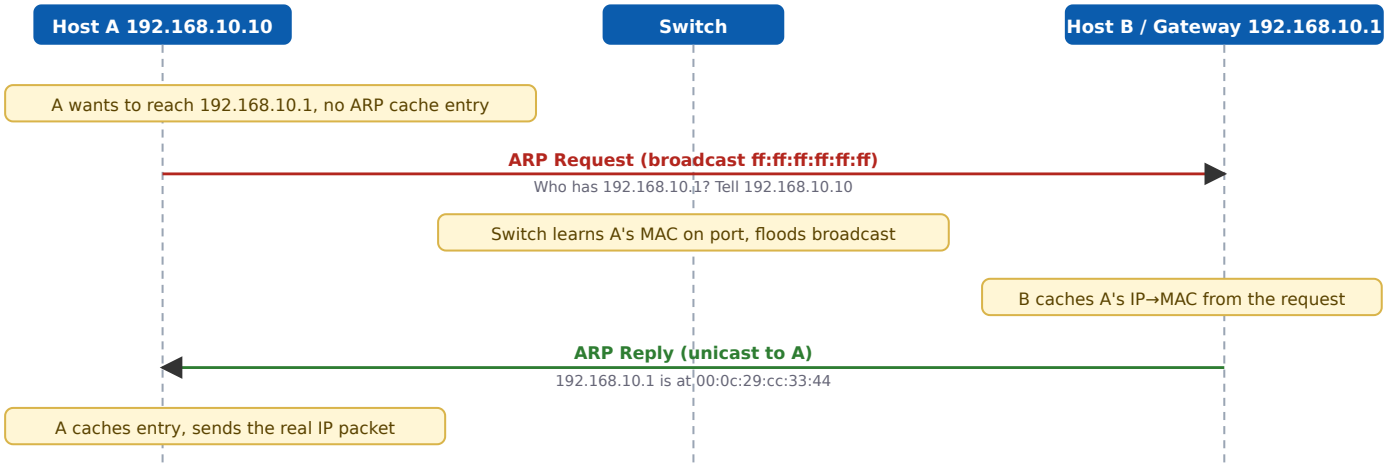
How hosts find each other on a LAN

2.1 Theory

- **MAC address:** 48-bit, written as 6 bytes hex (00:50:56:aa:11:22). First 3 bytes = OUI (vendor). Broadcast = `ff:ff:ff:ff:ff:ff`. IPv4 multicast MACs start with `01:00:5e`.
- **Ethernet II frame:** Dst MAC (6) | Src MAC (6) | EtherType (2) | Payload (46-1500) | FCS (4). Standard MTU is 1500 bytes; with VLAN tag the header grows by 4 bytes.
- **Switch behaviour:** learns *source* MAC per port, forwards by *destination* MAC. Unknown unicast, broadcast and multicast are flooded within the VLAN. Aging time is 300 s by default.
- **ARP (RFC 826):** maps IPv4 to MAC. Request is broadcast, reply is unicast. Result is cached (Windows ~ minutes, Cisco 4 hours by default).
- **Gratuitous ARP:** host announces its own IP/MAC (sender IP = target IP). Used for duplicate IP detection and failover (HSRP/VRRP, firewall HA) to update neighbours' caches.
- **Proxy ARP:** router answers ARP on behalf of a remote host. Can hide misconfigured masks or gateways.

2.2 ARP flow

Same-subnet ARP. For a remote destination, the host ARPs for the default gateway, not the destination.



2.3 Wireshark: ARP request and reply

arp.pcapng - Wireshark

Apply a display filter ... `arp`

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	00:50:56:aa:11:22	Broadcast	ARP	42	Who has 192.168.10.1? Tell 192.168.10.10
2	0.000312	00:0c:29:cc:33:44	00:50:56:aa:11:22	ARP	60	192.168.10.1 is at 00:0c:29:cc:33:44

```
▸ Frame 1: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface eth0, id 0
▾ Ethernet II, Src: 00:50:56:aa:11:22, Dst: ff:ff:ff:ff:ff:ff (1)
      Destination: ff:ff:ff:ff:ff:ff (Broadcast)
      Source: 00:50:56:aa:11:22
      Type: ARP (0x0806) (2)
▾ Address Resolution Protocol (request) (3)
      Hardware type: Ethernet (1)
      Protocol type: IPv4 (0x0800)
      Hardware size: 6    Protocol size: 4
      Opcode: request (1) (4)
      Sender MAC address: 00:50:56:aa:11:22 (5)
      Sender IP address: 192.168.10.10
      Target MAC address: 00:00:00:00:00:00 (6)
      Target IP address: 192.168.10.1 (7)
```

```
0000  ff ff ff ff ff ff 00 50  56 aa 11 22 08 06 00 01  .......PV.."....
0010  08 00 06 04 00 01 00 50  56 aa 11 22 c0 a8 0a 0a  .......PV.."....
0020  00 00 00 00 00 00 c0 a8  0a 01                    ..........
```

#	What it denotes
1	**Destination = ff:ff:ff:ff:ff:ff.** A request is a broadcast, so every host in the VLAN receives it. A switch floods it out of all ports in that VLAN.
2	**EtherType 0x0806** tells the receiver the payload is ARP, not IPv4 (0x0800).
3	ARP has no IP header at all; it rides directly on Ethernet, which is why routers never forward ARP.
4	**Opcode 1 = request, 2 = reply.**
5	Sender fields let every receiver learn the asker's mapping.
6	**Target MAC all zeros** = the unknown being asked for.
7	The IP being resolved. In the reply, Wireshark shows the answer in the Info column: *192.168.10.1 is at ...*

Figure 2.1: ARP request (packet 1 selected). The hex pane shows the same 42 bytes: ff.. (dst) | 00 50 56.. (src) | 08 06 (type) | ARP body.

Reading ARP problems in a capture

arp-problems.pcapng - Wireshark

Apply a display filter ... `arp.duplicate-address-detected || arp.opcode==1`

No.	Time	Source	Destination	Protocol	Len	Info
10	5.000000	00:50:56:aa:11:22	Broadcast	ARP	42	Who has 192.168.10.1? Tell 192.168.10.10
11	6.000412	00:50:56:aa:11:22	Broadcast	ARP	42	Who has 192.168.10.1? Tell 192.168.10.10
12	7.001001	00:50:56:aa:11:22	Broadcast	ARP	42	Who has 192.168.10.1? Tell 192.168.10.10
20	12.300000	00:0c:29:cc:33:44	Broadcast	ARP	60	192.168.10.1 is at 00:0c:29:cc:33:44 (duplicate use of 192.168.10.…
21	12.300410	00:50:56:99:99:99	Broadcast	ARP	60	192.168.10.1 is at 00:50:56:99:99:99 (duplicate use of 192.168.10.…

```
▾ Expert Info (Warning/Sequence): Duplicate IP address configured (192.168.10.1) (1)
      [Duplicate IP address configured (192.168.10.1)]
      [Frame showing earlier use of IP address: 21]
      [Seconds since earlier frame seen: 0]
```

#	What it denotes
1	**Duplicate IP:** two different MACs answer for one IP. Users see intermittent connectivity as ARP caches flip between the two MACs. Fix: find the rogue MAC via `show mac address-table address 0050.5699.9999`, trace the port, correct the config.

Figure 2.2: Top: repeated ARP requests with no reply (host down, wrong VLAN, port blocked). Bottom: duplicate IP detection.

2.4 Commands

```
# Windows / Linux host
arp -a                      ! show ARP cache
arp -d *                    ! clear cache (Windows, admin)
ip neigh show               ! Linux neighbour table

# Cisco switch / router
show mac address-table                     ! where is each MAC learned
show mac address-table address 0050.56aa.1122
show ip arp                                ! router ARP table (IP, age, MAC, interface)
clear ip arp 192.168.10.10

# Palo Alto
show arp all
show arp ethernet1/2
```

2.5 Scenario questions and troubleshooting

Scenario	How to troubleshoot
Host cannot ping gateway; `arp -a` shows *incomplete / no entry* for the gateway	Capture on host: repeated ARP requests, no reply. Check VLAN on access port, trunk allowed list, gateway interface up/up, port-security shutdown, DAI dropping, STP blocking.
Users randomly lose network, gateway pings drop every few minutes	Possible duplicate IP or two devices claiming the gateway IP. Look for ARP replies from two MACs in one capture; verify with `show ip arp` on router.
How does a switch know which port a MAC is on?	Learns the source MAC on ingress; stores MAC, VLAN, port. Unknown destinations are flooded.
What happens if the MAC table fills up (CAM overflow)?	Switch floods unknown unicast like a hub, allowing sniffing. Mitigate with port-security max MAC and DHCP snooping/DAI.
What is ARP spoofing and how do you stop it?	Attacker sends forged ARP replies to poison caches (man-in-the-middle). Mitigate with Dynamic ARP Inspection using DHCP snooping bindings, port security, static ARP for critical hosts.
Why does a gratuitous ARP appear after an HSRP failover?	The new active router announces the virtual IP / virtual MAC so switches relearn the MAC on the new port.

3. IP, ICMP, ping and traceroute

Layer 3 addressing and its diagnostic protocol

3.1 Theory

- **IPv4 header (20 bytes minimum):** Version, IHL, DSCP/ECN, Total Length, Identification, Flags (DF, MF), Fragment Offset, **TTL**, **Protocol** (1 ICMP, 6 TCP, 17 UDP, 47 GRE, 50 ESP, 89 OSPF), Header Checksum, Src IP, Dst IP.
- **MTU / fragmentation:** Ethernet MTU 1500. If a packet is larger than the link MTU and DF is set, the router drops it and sends ICMP type 3 code 4 (Fragmentation needed). Broken PMTUD is a common cause of "small pages load, big transfers hang", especially over VPN/IPSec (overhead reduces usable MSS).
- **ICMP:** control and error messages, carried inside IP (protocol 1). Not a transport protocol and has no ports.
- **TTL default:** Windows 128, Linux/macOS 64, Cisco 255. You can estimate hops from the reply TTL.

ICMP type/code	Meaning	Seen when
0 / 0	Echo reply	Ping response
8 / 0	Echo request	Ping
3 / 0	Network unreachable	Router has no route
3 / 1	Host unreachable	Router cannot ARP for the host
3 / 3	Port unreachable	UDP port closed (also how traceroute detects the destination)
3 / 4	Fragmentation needed, DF set	MTU problem
3 / 13	Administratively prohibited	ACL or firewall rejecting with ICMP
11 / 0	TTL exceeded in transit	Traceroute hop / routing loop
5	Redirect	Better next-hop exists on the same segment

3.2 Wireshark: ping

ping.pcapng - Wireshark

Apply a display filter ... icmp

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	192.168.10.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=1/256, ttl=128 (reply in 2)
2	0.018204	8.8.8.8	192.168.10.10	ICMP	74	Echo (ping) reply id=0x0001, seq=1/256, ttl=117 (request in 1)
3	1.003011	192.168.10.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=2/512, ttl=128 (reply in 4)
4	1.021890	8.8.8.8	192.168.10.10	ICMP	74	Echo (ping) reply id=0x0001, seq=2/512, ttl=117 (request in 3)

```
▸ Frame 1: 74 bytes on wire (592 bits)
▸ Ethernet II, Src: 00:50:56:aa:11:22, Dst: 00:0c:29:cc:33:44 (1)
▾ Internet Protocol Version 4, Src: 192.168.10.10, Dst: 8.8.8.8
      0100 .... = Version: 4
      .... 0101 = Header Length: 20 bytes (5)
      Total Length: 60
      Flags: 0x0000 (DF not set)
      Time to Live: 128 (2)
      Protocol: ICMP (1) (3)
      Source Address: 192.168.10.10
      Destination Address: 8.8.8.8
▾ Internet Control Message Protocol (4)
      Type: 8 (Echo (ping) request) (5)
      Code: 0
      Checksum: 0x4d5a [correct]
      Identifier (BE): 1 (0x0001) (6)
      Sequence Number (BE): 1 (0x0001) (6)
      [Response frame: 2]
    ▸ Data (32 bytes)
```

#	What it denotes
1	Destination MAC is the **gateway's MAC** because 8.8.8.8 is remote (see chapter 1/11).
2	TTL 128 on the request suggests a Windows sender. The reply TTL (117) = the responder's initial TTL minus the hops crossed on the way back. Only an estimate, since initial TTLs differ (64, 128, 255).
3	IP protocol number 1 = ICMP.
4	ICMP has no ports. It uses type and code.
5	Type 8 = request; reply is type 0.
6	Identifier and sequence number pair requests with replies. Missing sequence numbers in a capture = packet loss.

Figure 3.1: Successful ping. Packet 1 selected; Wireshark links request and reply ("reply in 2").

3.3 Wireshark: traceroute and ICMP errors

traceroute.pcapng - Wireshark

Apply a display filter ... `icmp.type==11 || icmp.type==3 || icmp`

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000	192.168.10.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=1, ttl=1 (no response found!)
2	0.001	192.168.10.1	192.168.10.10	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
3	0.004	192.168.10.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=2, ttl=2 (no response found!)
4	0.012	10.0.0.1	192.168.10.10	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
5	0.015	192.168.10.10	8.8.8.8	ICMP	74	Echo (ping) request id=0x0001, seq=3, ttl=3 (reply in 6)
6	0.031	8.8.8.8	192.168.10.10	ICMP	74	Echo (ping) reply id=0x0001, seq=3, ttl=118
9	3.200	192.168.10.10	10.9.9.9	ICMP	74	Echo (ping) request id=0x0001, seq=7, ttl=128 (no response found!)
10	3.201	192.168.10.1	192.168.10.10	ICMP	70	Destination unreachable (Host unreachable)
11	4.100	192.168.10.10	172.16.1.50	ICMP	74	Echo (ping) request ...
12	4.101	192.168.10.1	192.168.10.10	ICMP	70	Destination unreachable (Communication administratively filtered)

```
▸ Frame 2: 70 bytes on wire
▾ Internet Protocol Version 4, Src: 192.168.10.1, Dst: 192.168.10.10
      Source Address: 192.168.10.1 (1)
▾ Internet Control Message Protocol
      Type: 11 (Time-to-live exceeded) (2)
      Code: 0 (Time to live exceeded in transit) (3)
      Checksum: 0xf4ff [correct]
▾ Internet Protocol Version 4, Src: 192.168.10.10, Dst: 8.8.8.8 (4)
      Time to Live: 1
▸ Internet Control Message Protocol (original request, quoted)
```

#	What it denotes
1	The **source of the error is the router that dropped the packet**. That is how traceroute learns each hop's IP.
2	Type 11 = TTL exceeded. Seeing this at the end repeatedly with the same two routers = routing loop.
3	Code 0 = in transit (code 1 would be fragment reassembly).
4	The ICMP error quotes the original IP header + first 8 bytes. Use this to identify which flow caused it.

Figure 3.2: Windows tracert uses ICMP echo with increasing TTL (Linux traceroute uses UDP by default). Packets 9-12 show host-unreachable (router could not ARP the destination) and administratively filtered (ACL/firewall reject).

3.4 Commands

```
ping 8.8.8.8 -t                 ! Windows continuous (Ctrl+C)
ping 10.1.1.50 -f -l 1472       ! Windows DF set + size to test MTU 1500 (1472+28)
tracert -d 8.8.8.8              ! no DNS lookups
traceroute -n 8.8.8.8           ! Linux (UDP), -I for ICMP
pathping 8.8.8.8

! Cisco: extended ping / source selection
ping 10.1.1.50 source 192.168.10.1 repeat 100 size 1500 df-bit
traceroute 10.1.1.50
```

3.5 Scenario questions

Question	Answer outline
Ping gives "Request timed out". Is the host down?	Not necessarily. ICMP may be blocked by firewall/host policy, return route may be missing (asymmetry), or ARP failed. Test with a TCP port, check both directions, capture on both sides.
"Destination host unreachable" vs "Request timed out"?	Unreachable: some device (often the last-hop router, or local host) replied with ICMP unreachable. Timed out: no reply at all (drop, filter, missing return route).
How does traceroute work?	Sends probes with TTL 1,2,3...; each router decrements, drops at 0, returns ICMP Time Exceeded revealing its IP. Destination replies with echo reply (ICMP mode) or port unreachable (UDP mode). Stars mean the hop does not respond to probes, which does not always mean a fault.
Large file transfers hang over VPN but ping works	MTU/MSS issue. Test ping with DF and size; reduce MSS (TCP adjust-mss on the firewall tunnel interface or clamp), allow ICMP type 3 code 4.
Traceroute shows the same two hops repeating	Routing loop. Check routing tables on both devices (conflicting static routes, redistribution, floating static with wrong AD).

4. TCP and UDP

Transport layer: handshakes, flags, loss, resets

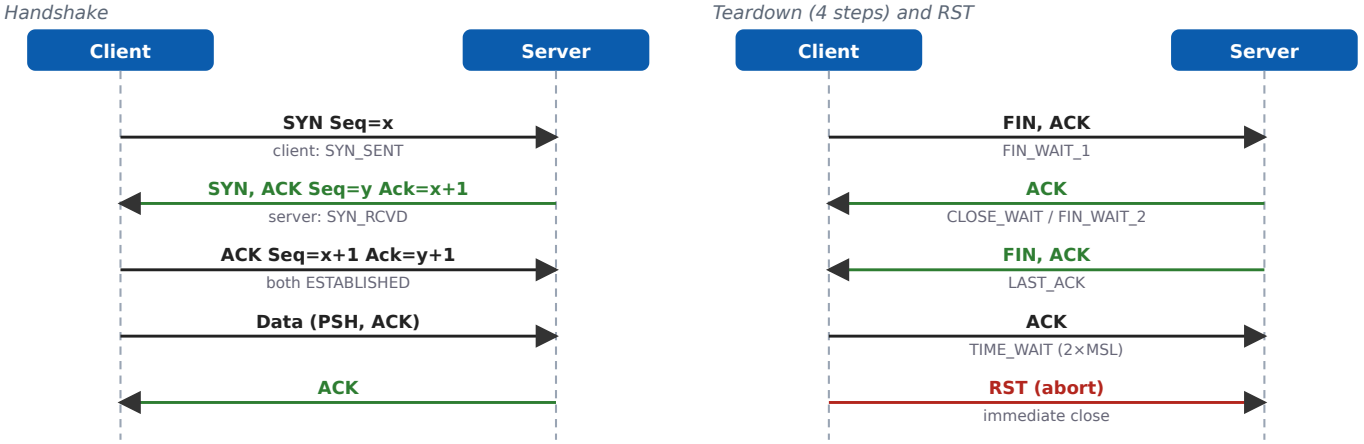
4.1 Theory

	TCP	UDP
Connection	Connection-oriented (3-way handshake)	Connectionless
Reliability	Sequence numbers, ACKs, retransmission, ordering	None (app must handle)
Flow control	Window size, congestion control	None
Header	20-60 bytes	8 bytes (src, dst, length, checksum)
Used by	HTTP/S, SSH, SMTP, BGP (179), FTP, RDP	DNS, DHCP, SNMP, syslog, NTP, TFTP, IKE (500/4500), VoIP/RTP

TCP header fields to know

Field	Meaning
Source / Dest port	16-bit. Client uses an ephemeral port (49152-65535 typical), server a well-known port.
Sequence number	Byte position of the first data byte in this segment. Initial value (ISN) is random; Wireshark shows relative numbers (starts at 0).
Acknowledgment number	Next byte the sender expects to receive. Ack = Seq + data length (SYN and FIN count as 1).
Flags	SYN start, ACK acknowledge, FIN graceful close, RST abort, PSH push to app, URG urgent.
Window size	Receiver's free buffer. Window scale option multiplies it (value × 2^scale).
Options	MSS (max segment size, 1460 on 1500 MTU), Window Scale, SACK permitted, Timestamps.

4.2 Three-way handshake and teardown



TCP state	Meaning	Seen on netstat when
LISTEN	Server waiting	Service running on port
SYN_SENT	Client sent SYN, waiting	Many = destination not answering
SYN_RECEIVED	Server got SYN, sent SYN-ACK	Many = SYN flood or return path broken
ESTABLISHED	Connection up	Normal
CLOSE_WAIT	Peer closed, local app has not	Many = application bug (not closing sockets)
TIME_WAIT	Waiting 2×MSL after close	Normal on busy clients

4.3 Wireshark: successful handshake

tcp-handshake.pcapng - Wireshark

Apply a display filter ... `tcp.stream eq 0`

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	192.168.10.10	10.1.1.50	TCP	66	51514 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
2	0.001850	10.1.1.50	192.168.10.10	TCP	66	443 → 51514 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_P…
3	0.001902	192.168.10.10	10.1.1.50	TCP	54	51514 → 443 [ACK] Seq=1 Ack=1 Win=262656 Len=0
4	0.002410	192.168.10.10	10.1.1.50	TLSv1.3	571	Client Hello
5	0.004120	10.1.1.50	192.168.10.10	TCP	54	443 → 51514 [ACK] Seq=1 Ack=518 Win=64768 Len=0

```
▸ Frame 1: 66 bytes on wire (528 bits)
▸ Ethernet II, Src: 00:50:56:aa:11:22, Dst: 00:0c:29:cc:33:44
▸ Internet Protocol Version 4, Src: 192.168.10.10, Dst: 10.1.1.50
▾ Transmission Control Protocol, Src Port: 51514, Dst Port: 443, Seq: 0, Len: 0
      Source Port: 51514 (1)
      Destination Port: 443 (1)
      [Stream index: 0] (2)
      [Conversation completeness: Complete, WITH_DATA (31)]
      [TCP Segment Len: 0]
      Sequence Number: 0    (relative sequence number) (3)
      Sequence Number (raw): 2917034210
      Acknowledgment Number: 0
      1000 .... = Header Length: 32 bytes (8)
      Flags: 0x002 (SYN) (4)
          .... .... ..1. = Syn: Set
      Window: 64240 (5)
      Checksum: 0x8f3a [unverified]
    ▾ Options: (12 bytes) (6)
          TCP Option - Maximum segment size: 1460 bytes
          TCP Option - No-Operation (NOP)
          TCP Option - Window scale: 8 (multiply by 256)
          TCP Option - SACK permitted
```

#	What it denotes
1	Client picks an **ephemeral source port** (51514); destination 443 = HTTPS. The pair of IP:port on both sides identifies the connection (5-tuple).
2	Wireshark numbers each connection (`tcp.stream`); right-click → Follow → TCP Stream.
3	**Relative** Seq=0 (Wireshark default). Raw ISN is random. Reply has Ack=1 because SYN consumes one sequence number.
4	Only **SYN** set = connection request. SYN+ACK from server, then a pure ACK from the client completes the handshake.
5	Window advertised in SYN is not scaled; after scaling (×256) packet 3 shows 262656.
6	Options appear only in SYN/SYN-ACK. **MSS 1460** = 1500 MTU − 20 IP − 20 TCP. If one side lacks Window Scale or SACK, throughput suffers. Firewalls rewriting options can break performance.

Figure 4.1: Normal TCP handshake followed by a TLS ClientHello (packet 4). The three green-light signs: SYN, SYN-ACK, ACK within a few milliseconds.

4.4 Wireshark: when the handshake fails

(a) SYN sent, nothing comes back

syn-no-reply.pcapng - Wireshark

Apply a display filter ... `tcp.flags.syn==1 && tcp.flags.ack==0 || tcp.analysis.retransmission`

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	192.168.10.10	10.1.1.50	TCP	66	51520 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
2	1.003114	192.168.10.10	10.1.1.50	TCP	66	[TCP Retransmission] 51520 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1…
3	3.011070	192.168.10.10	10.1.1.50	TCP	66	[TCP Retransmission] 51520 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1…
4	7.027400	192.168.10.10	10.1.1.50	TCP	66	[TCP Retransmission] 51520 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1…

```
▾ Transmission Control Protocol, Src Port: 51520, Dst Port: 443, Seq: 0, Len: 0
    ▸ Flags: 0x002 (SYN)
    ▾ [SEQ/ACK analysis] (1)
        ▾ [TCP Analysis Flags]
            [Expert Info (Note/Sequence): This frame is a (suspected) retransmission] (2)
          [The RTO for this segment was: 1.003114000 seconds] (3)
          [RTO based on delta from frame: 1]
```

#	What it denotes
1	Black rows with red text = Wireshark Expert Info flagged a problem.
2	Same SYN resent: nobody answered. The SYN is leaving this host, but no SYN-ACK arrives.
3	Intervals double (1 s, 2 s, 4 s on Linux; 3 s then 6 s on Windows). Typical causes: **firewall silently dropping**, missing route or return route, wrong IP, host down, ACL, upstream filtering. Capture on the server side too: if the SYN never arrives, the drop is in the path; if it arrives and SYN-ACK leaves but never returns, suspect asymmetric routing or a firewall on the return path.

Figure 4.2: Classic 'timeout' signature. Compare with 4.3 below, where an RST means something answered actively.

(b) Port closed or connection reset

rst.pcapng - Wireshark

Apply a display filter ... `tcp.flags.reset==1`

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	192.168.10.10	10.1.1.50	TCP	66	51522 → 8080 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
2	0.001420	10.1.1.50	192.168.10.10	TCP	60	8080 → 51522 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
10	9.100000	192.168.10.10	10.1.1.50	TCP	571	51530 → 443 [PSH, ACK] Seq=1 Ack=1 Win=262656 Len=517
11	9.101800	10.1.1.50	192.168.10.10	TCP	60	443 → 51530 [RST, ACK] Seq=1 Ack=518 Win=0 Len=0

```
▾ Internet Protocol Version 4, Src: 10.1.1.50, Dst: 192.168.10.10
      Time to Live: 64 (1)
▾ Transmission Control Protocol, Src Port: 8080, Dst Port: 51522, Seq: 1, Ack: 1, Len: 0
    ▾ Flags: 0x014 (RST, ACK) (2)
          .... .... .1.. = Reset: Set
          .... ...1 .... = Acknowledgment: Set
      Window: 0
```

#	What it denotes
1	**Who sent the RST?** Compare this TTL with TTL of normal packets from the same server. If the server's SYN-ACKs arrive with TTL 128 but the RST has TTL 64 or 255, a middlebox (firewall, IPS, load balancer) probably injected it.
2	**RST right after SYN** = port not listening or an explicit reject policy. **RST in the middle of a session** (packet 11) = application abort, firewall/IPS policy, session timeout expired on a firewall, or half-open state after a reboot. Cisco ACL deny normally drops silently; Palo Alto *reset-client/reset-server/reset-both* actions send RST.

Figure 4.3: Red row with white text = TCP RST. Immediate refusal (packet 2) vs. mid-session reset (packet 11).

4.5 Wireshark: packet loss, duplicate ACKs, zero window

loss-and-window.pcapng - Wireshark

Apply a display filter ... `tcp.analysis.flags`

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	192.168.10.10	10.1.1.50	TCP	15…	51540 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=1448
2	0.000020	192.168.10.10	10.1.1.50	TCP	15…	51540 → 80 [ACK] Seq=1449 Ack=1 Win=64256 Len=1448 ← lost on the w…
3	0.000040	192.168.10.10	10.1.1.50	TCP	15…	51540 → 80 [ACK] Seq=2897 Ack=1 Win=64256 Len=1448
4	0.000060	192.168.10.10	10.1.1.50	TCP	15…	51540 → 80 [ACK] Seq=4345 Ack=1 Win=64256 Len=1448
5	0.000080	192.168.10.10	10.1.1.50	TCP	15…	51540 → 80 [ACK] Seq=5793 Ack=1 Win=64256 Len=1448
6	0.021000	10.1.1.50	192.168.10.10	TCP	60	80 → 51540 [ACK] Seq=1 Ack=1449 Win=64128 Len=0
7	0.021900	10.1.1.50	192.168.10.10	TCP	66	[TCP Dup ACK 6#1] 80 → 51540 [ACK] Seq=1 Ack=1449 Win=64128 Len=0 …
8	0.022100	10.1.1.50	192.168.10.10	TCP	66	[TCP Dup ACK 6#2] 80 → 51540 [ACK] Seq=1 Ack=1449 Win=64128 Len=0 …
9	0.022300	10.1.1.50	192.168.10.10	TCP	66	[TCP Dup ACK 6#3] 80 → 51540 [ACK] Seq=1 Ack=1449 Win=64128 Len=0 …
10	0.022500	192.168.10.10	10.1.1.50	TCP	15…	[TCP Fast Retransmission] 51540 → 80 [ACK] Seq=1449 Ack=1 Win=6425…
11	0.043700	10.1.1.50	192.168.10.10	TCP	60	80 → 51540 [ACK] Seq=1 Ack=7241 Win=64128 Len=0
12	2.100000	10.1.1.50	192.168.10.10	TCP	60	[TCP ZeroWindow] 80 → 51540 [ACK] Seq=1 Ack=7241 Win=0 Len=0

```
▾ Transmission Control Protocol, Src Port: 80, Dst Port: 51540, Seq: 1, Ack: 1449, Len: 0
    Acknowledgment Number: 1449    (relative ack number) (1)
  ▾ Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), SACK
    ▾ TCP Option - SACK 2897-4345 (2)
        Left Edge = 2897
        Right Edge = 4345
  ▾ [SEQ/ACK analysis]
    ▾ [TCP Analysis Flags]
        [Expert Info (Note/Sequence): Duplicate ACK (#1)] (3)
      [This is a TCP duplicate ack]
      [Duplicate ACK #: 1]
      [Duplicate to the ACK in frame: 6]
```

#	What it denotes
1	Receiver keeps saying **"I still need byte 1449"**, because that segment (packet 2) never arrived.
2	**SACK blocks** tell the sender what did arrive (2897-4345 etc.), so only the missing segment is resent.
3	**3 duplicate ACKs trigger fast retransmit** (packet 10) without waiting for a timeout (RTO). A plain *[TCP Retransmission]* after a long pause means RTO expired, which hurts throughput much more.
4	**[TCP ZeroWindow]** (packet 12): the receiver's buffer is full. This is a slow receiver/application issue, not network loss. A burst of *TCP Window Update* follows when it recovers. Persistent Window Full or ZeroWindow points to the server or client app.

Figure 4.4: Loss recovery. Where were the packets lost? Capture at both ends: packets present at the sender but absent at the receiver were lost in between (check interface errors, queue drops, duplex, firewall, ISP).

Symptom in capture	Likely meaning	Where to look
SYN, SYN retransmissions, no SYN-ACK	Blocked or unreachable	Route, ACL/firewall drop, NAT, security policy, server down
SYN then immediate RST	Port closed / rejected	Service status, firewall reject rule, load-balancer
Handshake OK, then retransmissions	Loss or MTU problem	Interface errors, duplex, congestion, MTU/MSS, VPN overhead
Many Dup ACKs / retransmissions	Packet loss	Capture both ends, find hop
ZeroWindow	Slow receiver	Server CPU/app, client app
High time between request and response	Server/application latency	`tcp.time_delta`, `tcp.analysis.ack_rtt`
RST in middle of data	Reset by device or app	TTL of RST, firewall logs, session timeout
FIN from one side only, other side stays	Half-closed or app not closing	netstat CLOSE_WAIT

INTERVIEW ANGLE

"How do you decide whether the problem is the network or the application?" Look at the handshake. If SYN-ACK returns quickly, the network path and port are fine. Then measure the time between the request (PSH) and the server's response: a delay there is application/server time, not network. Packet loss shows as retransmissions and dup ACKs.

4.6 Scenario questions

Question	Answer outline
What is the difference between TCP and UDP, and why does DNS use UDP?	TCP reliable/connection-oriented; UDP lightweight. DNS queries are small and single exchange, so UDP avoids handshake overhead. TCP is used for large responses (>512 B without EDNS), zone transfers, DNSSEC.
What is a SYN flood?	Attacker sends many SYNs and never completes the handshake, filling the server's half-open queue. Mitigated with SYN cookies, firewall SYN-flood protection (Palo Alto Zone/DoS protection), rate limiting.
Why does the SYN-ACK have Ack=1?	SYN consumes one sequence number, so the next expected byte is x+1.

Application slow only for some users	Compare captures, check RTT, loss, MTU, window scale, DNS time, and path (asymmetric/VPN). Find what differs between affected and unaffected users.
What is TCP MSS clamping?	Firewall/router rewrites the MSS in SYN packets to a lower value so tunnels (IPSec/GRE) do not need fragmentation. Example: Cisco `ip tcp adjust-mss 1360` .
Session drops after ~1 hour idle through the firewall	Firewall session timeout shorter than the app's idle period (TCP default is 3600 s on Palo Alto). Fix with application keepalives, longer custom timeout for that app, or TCP keepalive.

5. DNS and DHCP

Name resolution and address assignment

5.1 DNS theory

- **Flow:** app → local cache / hosts file → configured resolver (recursive) → root → TLD (.com) → authoritative server → answer cached for TTL.
- **Ports:** UDP/53 for queries, TCP/53 for large replies and zone transfers (AXFR).
- **Record types:** A (name→IPv4), AAAA (IPv6), CNAME (alias), MX (mail), NS (name server), PTR (reverse), SOA, TXT, SRV.
- **Response codes:** NOERROR (0), FORMERR, SERVFAIL (2, server problem), NXDOMAIN (3, name does not exist), REFUSED (5, policy).

5.2 Wireshark: DNS query, response and NXDOMAIN

dns.pcapng - Wireshark

Apply a display filter ... dns

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	192.168.10.10	192.168.10.5	DNS	74	Standard query 0x3a7b A app.corp.local
2	0.012300	192.168.10.5	192.168.10.10	DNS	90	Standard query response 0x3a7b A app.corp.local A 172.16.1.50
3	4.500000	192.168.10.10	192.168.10.5	DNS	74	Standard query 0x4c10 A appp.corp.local
4	4.511000	192.168.10.5	192.168.10.10	DNS	133	Standard query response 0x4c10 No such name A appp.corp.local SOA …
5	9.000000	192.168.10.10	10.99.99.99	DNS	74	Standard query 0x4c11 A app.corp.local
6	10.000000	192.168.10.10	10.99.99.99	DNS	74	[Retransmission] Standard query 0x4c11 A app.corp.local

```
▸ Frame 2: 90 bytes on wire
▸ Ethernet II
▸ Internet Protocol Version 4, Src: 192.168.10.5, Dst: 192.168.10.10
▾ User Datagram Protocol, Src Port: 53, Dst Port: 53124 (1)
▾ Domain Name System (response)
      Transaction ID: 0x3a7b (2)
    ▾ Flags: 0x8180 Standard query response, No error (3)
          1... .... .... .... = Response: Message is a response
          .000 0... .... .... = Opcode: Standard query (0)
          .... .0.. .... .... = Authoritative: Server is not an authority for domain
          .... ...1 .... .... = Recursion desired: Do query recursively
          .... .... 1... .... = Recursion available: Server can do recursive queries
          .... .... .... 0000 = Reply code: No error (0) (4)
      Questions: 1    Answer RRs: 1    Authority RRs: 0    Additional RRs: 0
    ▾ Queries
          app.corp.local: type A, class IN
    ▾ Answers (5)
        ▾ app.corp.local: type A, class IN, addr 172.16.1.50
              Time to live: 3600 (1 hour)
              Address: 172.16.1.50
      [Request In: 1]    [Time: 0.012300000 seconds] (6)
```

#	What it denotes
1	UDP port 53 on the server side; the client uses a random ephemeral port.
2	**Transaction ID** matches request to response (0x3a7b in packet 1 and 2).
3	Flags 0x8180 = response + recursion desired + recursion available. 0x0100 is a plain query.
4	**Reply code**: 0 No error. In packet 4 it is 3 (NXDOMAIN, typo or missing record). SERVFAIL (2) = resolver cannot get an answer upstream.
5	Answer section carries the record and its **TTL**. Low TTL means frequent lookups; stale DNS after a change = old TTL still cached.
6	**Response time**. Anything over ~100 ms for internal DNS deserves investigation. Packets 5-6: no reply and the client retries = DNS server unreachable (wrong server IP, firewall on UDP/53).

Figure 5.1: Successful lookup (1-2), NXDOMAIN (3-4, black row), and no response to a query (5-6).

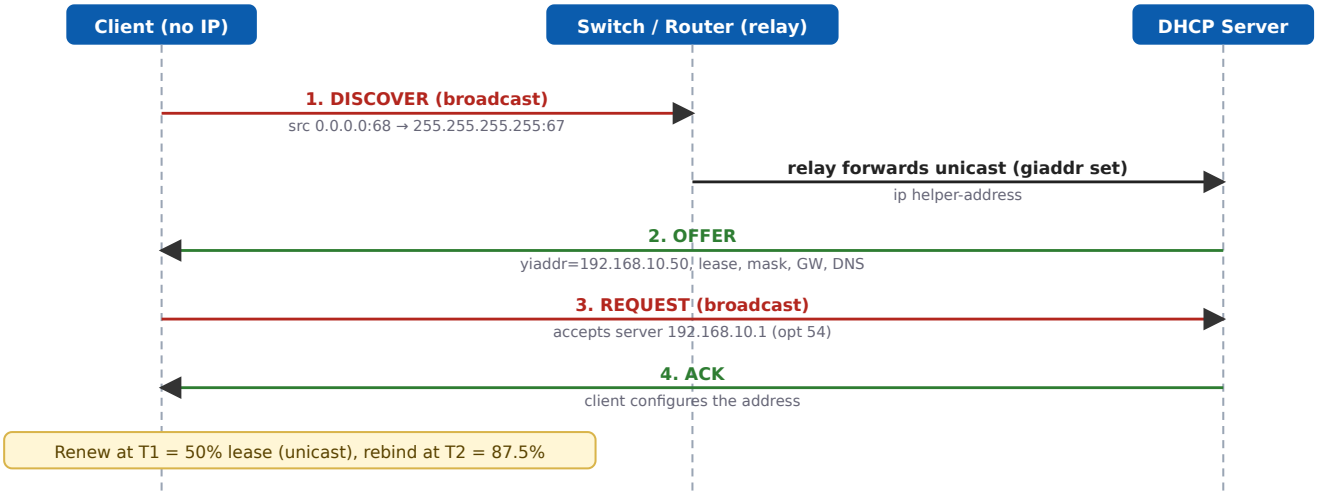
```
nslookup app.corp.local 192.168.10.5     ! query a specific server
nslookup -type=MX corp.local
dig @8.8.8.8 app.corp.local +short ; dig +trace example.com
ipconfig /displaydns ; ipconfig /flushdns   ! Windows cache
show ip dns view / ip name-server 192.168.10.5  ! Cisco client-side
```

DNS scenario	Troubleshooting
Internet works by IP but not by name	Check DNS server settings (DHCP option 6), nslookup against it, test UDP/53 reachability, firewall rules, resolver health.

Name resolves to old IP after migration	Cached record: check TTL, flush client cache, flush resolver, verify the authoritative record.
Internal names fail only over VPN	VPN client not given internal DNS server/suffix; split-DNS not configured; DNS traffic not allowed through tunnel policy.
Intermittent failures	Two DNS servers with different data or one is down; client alternates. Compare answers from each server.

5.3 DHCP theory and flow (DORA)

DHCP uses UDP: server port 67, client port 68



dhcp-dora.pcapng - Wireshark

Apply a display filter ... dhcp

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x5c1f2a3b
2	0.012500	192.168.10.1	192.168.10.50	DHCP	342	DHCP Offer - Transaction ID 0x5c1f2a3b
3	0.013100	0.0.0.0	255.255.255.255	DHCP	342	DHCP Request - Transaction ID 0x5c1f2a3b
4	0.020000	192.168.10.1	192.168.10.50	DHCP	342	DHCP ACK - Transaction ID 0x5c1f2a3b

```
▸ Frame 2: 342 bytes on wire
▸ Ethernet II, Src: 00:0c:29:cc:33:44, Dst: 00:50:56:aa:11:22
▸ Internet Protocol Version 4, Src: 192.168.10.1, Dst: 192.168.10.50
▸ User Datagram Protocol, Src Port: 67, Dst Port: 68
▾ Dynamic Host Configuration Protocol (Offer)
      Message type: Boot Reply (2)
      Transaction ID: 0x5c1f2a3b (1)
      Bootp flags: 0x0000 (Unicast)
      Client IP address: 0.0.0.0
      Your (client) IP address: 192.168.10.50 (2)
      Relay agent IP address: 0.0.0.0 (3)
      Client MAC address: 00:50:56:aa:11:22
      Magic cookie: DHCP
    ▾ Option: (53) DHCP Message Type (Offer) (4)
      Option: (54) DHCP Server Identifier (192.168.10.1) (5)
      Option: (51) IP Address Lease Time (28800s = 8 hours)
      Option: (1) Subnet Mask (255.255.255.0)
      Option: (3) Router: 192.168.10.1 (6)
      Option: (6) Domain Name Server: 192.168.10.5 (6)
      Option: (255) End
```

#	What it denotes
1	Same **Transaction ID** in all four packets ties the DORA together; use `dhcp.id == 0x5c1f2a3b` to filter.
2	**yiaddr** = the address being offered.
3	**giaddr** (relay agent IP). When the client and server are in different subnets, the router with *ip helper-address* fills this in. The server uses it to choose the correct scope. Zero here = server is on the same subnet.
4	Option 53 identifies the message type: Discover, Offer, Request, ACK, NAK, Decline, Release, Inform.
5	Server identifier. If two Offers come from different servers, you have a **rogue DHCP server**.
6	Options pushed to the client: gateway (3) and DNS (6). Wrong option here = clients with correct IPs but broken connectivity or name resolution.

Figure 5.2: DORA. Packet 2 (Offer) selected. Bootp flags 0x0000 = server replies unicast; 0x8000 = client asked for broadcast.

dhcp-fail.pcapng - Wireshark

Apply a display filter ... `dhcp || arp`

No.	Time	Source	Destination	Protocol	Len	Info
1	0.0	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7a1b0001
2	4.0	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7a1b0002
3	12.0	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7a1b0003
4	20.0	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0x7a1b0004 (no Offer received → cli…

▸ Dynamic Host Configuration Protocol (Discover)

Figure 5.3: Repeated Discover with no Offer. Capture on the DHCP server or relay to see if the Discover arrived.

5.4 Configuration (Cisco)

```
! Router/L3 switch as DHCP server
ip dhcp excluded-address 192.168.10.1 192.168.10.20
ip dhcp pool VLAN10
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 192.168.10.5
 lease 0 8
!
! Relay: on the gateway SVI, point to the central DHCP server
interface Vlan10
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 10.1.1.10
!
! DHCP snooping (stops rogue servers; builds binding table for DAI)
ip dhcp snooping
ip dhcp snooping vlan 10
interface g1/0/48
 ip dhcp snooping trust          ! uplink / real server port only

show ip dhcp binding ; show ip dhcp pool ; show ip dhcp conflict
show ip dhcp snooping binding
```

DHCP scenario	Troubleshooting
PC has 169.254.x.x (APIPA)	No Offer received. Check VLAN/port, trunk, helper-address on SVI, server reachability, scope exhausted, DHCP snooping dropping (untrusted uplink), server down.
Client in new VLAN gets no IP	Missing `ip helper-address` on the VLAN gateway, or no scope for that subnet on the server (server selects scope by giaddr).
Users get wrong gateway / 192.168.1.x addresses	Rogue DHCP (home router plugged in). Capture `dhcp.option.dhcp == 2` (Offers) and find two server IDs. Enable DHCP snooping.
Scope exhausted	Short lease with many clients, devices leaving. Shorten the lease, enlarge scope, clear stale bindings, check for DHCP starvation attack.
IP conflict warnings	Static IP inside DHCP range; exclude static addresses or investigate duplicate with ARP capture.

6. Subnetting and NAT/PAT

The calculations interviewers love, and address translation

6.1 Subnetting method that always works

1. Find the interesting octet (the one where the mask is not 255 or 0).
2. **Block size = 256 − mask value in that octet.**
3. Network = largest multiple of block size ≤ address value in that octet.
4. Broadcast = next network − 1. Usable range = network+1 to broadcast−1.
5. Hosts = $2^{\text{host bits}} - 2$.

CIDR	Mask	Block (octet)	Hosts	CIDR	Mask	Block	Hosts
/24	255.255.255.0	256	254	/28	255.255.255.240	16	14
/25	255.255.255.128	128	126	/29	255.255.255.248	8	6
/26	255.255.255.192	64	62	/30	255.255.255.252	4	2
/27	255.255.255.224	32	30	/31	255.255.255.254	2	2 (p2p, RFC 3021)
/23	255.255.254.0	2 (3rd)	510	/22	255.255.252.0	4 (3rd)	1022
/20	255.255.240.0	16 (3rd)	4094	/16	255.255.0.0	256 (2nd)	65534

Worked examples

Question	Working	Answer
192.168.10.77/26	Mask last octet 192, block 64. 77 lies in 64-127.	Network 192.168.10.64, broadcast .127, hosts .65-.126 (62)
172.16.45.200/20	Mask 255.255.240.0, block 16 in 3rd octet. 45 → 32.	Network 172.16.32.0, broadcast 172.16.47.255, hosts 4094
10.1.1.130/25	Block 128. 130 → 128.	Network 10.1.1.128, broadcast 10.1.1.255, 126 hosts
Same subnet? 192.168.10.30/27 and 192.168.10.40/27	Block 32: .30 in 0-31, .40 in 32-63.	No, different subnets (hosts need a router)
Summarise 10.1.0.0/24 - 10.1.3.0/24	Four /24s = two bits of the 3rd octet.	10.1.0.0/22
Smallest subnet for 50 hosts	$2^6-2 = 62 \geq 50$ → 6 host bits.	/26

VLSM example: carve 192.168.1.0/24

Need	Subnet	Range	Broadcast
100 hosts	192.168.1.0/25	1.1 - 1.126	1.127
50 hosts	192.168.1.128/26	1.129 - 1.190	1.191
25 hosts	192.168.1.192/27	1.193 - 1.222	1.223
10 hosts	192.168.1.224/28	1.225 - 1.238	1.239
Router link	192.168.1.240/30	1.241 - 1.242	1.243

TIP

Allocate the largest subnet first, aligned on its block boundary, and keep the remainder contiguous. Always verify there is no overlap.

Special ranges	Value
Private (RFC 1918)	10.0.0.0/8, 172.16.0.0/12 (172.16-172.31), 192.168.0.0/16
Loopback	127.0.0.0/8
APIPA / link-local	169.254.0.0/16
CGNAT	100.64.0.0/10
Multicast	224.0.0.0/4 (OSPF 224.0.0.5/.6, HSRPv1 224.0.0.2, EIGRP 224.0.0.10)
Documentation	192.0.2.0/24, 198.51.100.0/24, 203.0.113.0/24

6.2 Practice (answers at the end)

1. Network and broadcast of 192.168.5.100/27?
2. How many usable hosts in 10.0.0.0/21?
3. Is 172.16.17.5/20 in the same subnet as 172.16.31.250/20?
4. What mask gives at least 30 hosts and the most subnets from a /24?
5. Summarise 192.168.8.0/24 to 192.168.15.0/24.

Answers: 1) 192.168.5.96 / 192.168.5.127. 2) $2^{11}-2 = 2046$. 3) /20 block 16 in the 3rd octet: 17→16 and 31→16, both 172.16.16.0/20, so yes. 4) /27 (30 hosts). 5) 192.168.8.0/21.

6.3 NAT theory

Type	What it does	Use
Static NAT	One-to-one fixed mapping	Publishing a server (inside 10.1.1.50 ↔ public 203.0.113.50)
Dynamic NAT	Pool of public IPs, first come first served	Rare
PAT / NAT overload	Many inside to one public IP, distinguished by source port	Internet access for users (most common)
Destination NAT / port forward	Public IP:port → internal IP:port	Inbound services

Cisco terms: **inside local** (real private IP), **inside global** (public IP the world sees), **outside local/global** (remote host as seen inside/outside).

Stage	Src IP:port	Dst IP:port
Inside host sends	192.168.10.10:51514	93.184.216.34:443
After PAT on firewall/router	203.0.113.2:40001	93.184.216.34:443
Server reply	93.184.216.34:443	203.0.113.2:40001
Returned to host (un-NAT)	93.184.216.34:443	192.168.10.10:51514

```
! Cisco PAT
interface g0/0 ; ip nat outside
interface g0/1 ; ip nat inside
access-list 10 permit 192.168.10.0 0.0.0.255
ip nat inside source list 10 interface g0/0 overload
! Static NAT (server publishing)
ip nat inside source static 10.1.1.50 203.0.113.50
show ip nat translations ; show ip nat statistics ; clear ip nat translation *
```

WATCH OUT
Palo Alto rule: security policy is evaluated using the **pre-NAT IP addresses** but the **post-NAT destination zone**. Many failed server-publishing rules come from writing the policy with the translated address. Check with `test security-policy-match` and `test nat-policy-match`.

NAT scenario	Troubleshooting
Internal users cannot reach the internet after a change	Check inside/outside interface roles, ACL used for NAT, translation table, default route, firewall NAT policy and zones, session table for NAT source.
Published server unreachable from outside	Static/destination NAT present? Security policy matches pre-NAT destination and post-NAT zone? Server's default gateway points back through the same firewall (else asymmetric)? Upstream knows the public IP (proxy ARP / routing)?
Only some apps break through NAT	ALG-sensitive protocols (SIP, FTP, H.323). Check ALG/App-ID, or protocol embeds IPs in payload.
Port exhaustion	PAT limited to ~64k ports per public IP per destination. Add more public IPs to the pool, review long-lived sessions.

7. Switching: VLAN, trunk, STP, EtherChannel, security

Access and distribution layer operations

7.1 VLANs and trunking

- **VLAN** = one broadcast domain (usually one subnet). Splits a switch logically; broadcasts do not cross VLANs.
- **Access port:** carries one VLAN, frames untagged. **Trunk port:** carries many VLANs; frames tagged with 802.1Q (4 bytes: TPID 0x8100, PCP/priority 3 bits, DEI 1 bit, VLAN ID 12 bits = 1-4094).
- **Native VLAN:** the one VLAN sent untagged on a trunk (default 1). Must match on both ends; mismatch causes CDP warnings and leaks traffic between VLANs. Best practice: use an unused VLAN as native.
- **Voice VLAN:** phone tagged in the voice VLAN, PC untagged in the data VLAN, on the same access port.
- **Inter-VLAN routing:** router-on-a-stick (sub-interfaces with dot1q) or L3 switch SVIs (`interface vlan 10`). An SVI is up only if the VLAN exists and at least one up port is in it (or trunk carries it).
- **DTP:** auto-negotiates trunks. Disable with `switchport nonegotiate` and set modes explicitly.

```
vlan 10
 name USERS
vlan 20
 name SERVERS
interface g1/0/5
 description PC-Room1
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 spanning-tree bpduguard enable
interface g1/0/48
 description UPLINK-to-DIST
 switchport trunk encapsulation dot1q     ! older platforms only
 switchport mode trunk
 switchport trunk native vlan 999
 switchport trunk allowed vlan 10,20,999
 switchport nonegotiate
! Inter-VLAN on L3 switch
ip routing
interface vlan 10
 ip address 192.168.10.1 255.255.255.0
 no shutdown
! Router-on-a-stick
interface g0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
```

Wireshark: 802.1Q tagged frames on a trunk

trunk-capture.pcapng - Wireshark (SPAN/mirror of a trunk port)

Apply a display filter ... `vlan.id==20 (or: vlan)`

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	192.168.20.10	192.168.20.1	ICMP	78	Echo (ping) request id=0x0002, seq=1/256, ttl=128 (reply in 2)
2	0.000481	192.168.20.1	192.168.20.10	ICMP	78	Echo (ping) reply id=0x0002, seq=1/256, ttl=255 (request in 1)
3	0.250000	00:50:56:bb:77:88	Broadcast	ARP	64	Who has 192.168.30.1? Tell 192.168.30.25
4	0.500000	SW1	CDP/VTP/DTP/PAgP/…	CDP	391	Device ID: SW1 Port ID: GigabitEthernet1/0/48 (untagged = native V…

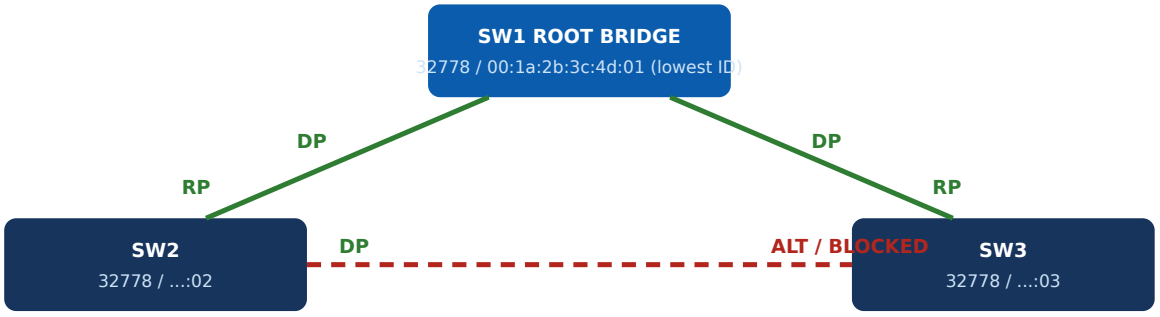
```
▸ Frame 1: 78 bytes on wire (624 bits)
▾ Ethernet II, Src: 00:50:56:aa:20:10, Dst: 00:0c:29:cc:20:01
      Type: 802.1Q Virtual LAN (0x8100) 1
▾ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20 2
      000. .... .... .... = Priority: Best Effort (default) (0) 3
      ...0 .... .... .... = DEI: Ineligible
      .... 0000 0001 0100 = ID: 20 4
      Type: IPv4 (0x0800)
▸ Internet Protocol Version 4, Src: 192.168.20.10, Dst: 192.168.20.1
▸ Internet Control Message Protocol
```

#	What it denotes
1	EtherType **0x8100** instead of 0x0800: this frame is tagged. The real EtherType follows after the tag.
2	The 4-byte 802.1Q header. Frame is 78 bytes instead of 74 (+4).
3	Priority bits (CoS), used for voice/QoS.
4	**VLAN ID 20**. Frame 3 is VLAN 30 broadcast ARP. If a VLAN does not appear in the capture, check the trunk allowed list and whether the VLAN exists on both switches. Untagged frames (packet 4) belong to the native VLAN.

Figure 7.1: Frames on a trunk. Note: many NICs strip VLAN tags before Wireshark sees them; capture via SPAN/mirror or enable tagging in the driver. Wireshark filter `vlan.id == 20`.

7.2 Spanning Tree Protocol

Why: redundant L2 links create loops. Ethernet frames have no TTL, so broadcast storms bring the network down in seconds. STP blocks redundant links logically and reopens them on failure.



Root has all ports Designated. Each non-root has one Root Port (best path to root). On each segment one Designated Port; the other end blocks.

Triangle topology: the SW2–SW3 link is blocked on SW3's side because SW2 has the lower bridge ID (tie-break after cost).

- **Bridge ID** = priority (default 32768, in steps of 4096) + extended system ID (VLAN number) + MAC. **Lowest wins root election.** Set root intentionally: `spanning-tree vlan 10 root primary` (priority 24576) or `priority 4096`.
- **Path cost** (short): 10 Mbps = 100, 100 Mbps = 19, 1 Gbps = 4, 10 Gbps = 2. Port selection tie-breakers: lowest root path cost → lowest sender bridge ID → lowest sender port ID.
- **Port roles:** Root (best path to root), Designated (forwards on a segment), Alternate/Blocked (backup path to root), Backup (RSTP, same switch).
- **802.1D timers:** hello 2 s, max age 20 s, forward delay 15 s. States: Blocking → Listening (15 s) → Learning (15 s) → Forwarding: 30-50 s convergence.
- **RSTP (802.1w):** states Discarding/Learning/Forwarding; proposal/agreement handshake converges in about 1-2 s. Edge ports (PortFast) skip delays. **Cisco default is PVST+ (one tree per VLAN)**, **Rapid-PVST+** is recommended: `spanning-tree mode rapid-pvst`. MST maps VLANs to few instances.

Feature	Purpose	Where
PortFast	Port goes straight to forwarding	Access ports to hosts only
BPDU Guard	Err-disable the port if a BPDU arrives	Access/PortFast ports (stops rogue switches)
BPDU Filter	Suppress BPDUs	Rarely; dangerous
Root Guard	Port ignores superior BPDUs (keeps root where you want it)	Distribution ports toward access

Loop Guard / UDLD	Protect against unidirectional links	Uplinks / fibre links
Storm control	Limit broadcast/multicast/unicast flood rate	Access ports

```
spanning-tree mode rapid-pvst
spanning-tree vlan 10,20 root primary          ! on the distribution switch
spanning-tree vlan 10,20 root secondary        ! on the redundant one
interface g1/0/5 ; spanning-tree portfast ; spanning-tree bpduguard enable
interface g1/0/10 ; spanning-tree guard root
storm-control broadcast level 1.0              ! percent of bandwidth
errdisable recovery cause bpduguard
errdisable recovery interval 300

show spanning-tree vlan 10
show spanning-tree summary
show spanning-tree interface g1/0/1 detail
show interfaces status err-disabled
```

```
SW3# show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    32778
             Address     001a.2b3c.4d01
             Cost        4                  <- path cost from this switch to root
             Port        1 (GigabitEthernet1/0/1)   <- root port
  Bridge ID  Priority    32778  (priority 32768 sys-id-ext 10)
             Address     001a.2b3c.4d03
Interface        Role Sts Cost      Prio.Nbr Type
Gi1/0/1          Root FWD 4         128.1    P2p
Gi1/0/2          Altn BLK 4         128.2    P2p      <- blocked redundant link
```

Wireshark: STP BPDUs

stp.pcapng - Wireshark

Apply a display filter ... stp

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	00:1a:2b:3c:4d:01	PVST+	STP	64	Conf. Root = 32768/10/00:1a:2b:3c:4d:01 Cost = 0 Port = 0x8001
2	0.000210	00:1a:2b:3c:4d:02	PVST+	STP	64	Conf. Root = 32768/10/00:1a:2b:3c:4d:01 Cost = 4 Port = 0x8002
3	2.000100	00:1a:2b:3c:4d:01	PVST+	STP	64	Conf. Root = 32768/10/00:1a:2b:3c:4d:01 Cost = 0 Port = 0x8001
4	2.000330	00:1a:2b:3c:4d:02	PVST+	STP	64	Conf. Root = 32768/10/00:1a:2b:3c:4d:01 Cost = 4 Port = 0x8002
5	7.400000	00:1a:2b:3c:4d:09	PVST+	STP	64	Conf. TC + Root = 4096/10/00:1a:2b:3c:4d:09 Cost = 0 Port = 0x8003…

```
▸ Frame 1: 64 bytes on wire (512 bits)
▾ IEEE 802.3 Ethernet
      Destination: PVST+ (01:00:0c:cc:cc:cd)  (VLAN 1 uses 01:80:c2:00:00:00)
      Source: 00:1a:2b:3c:4d:01
▸ Logical-Link Control (SNAP, Cisco PVSTP+ 0x010b)
▾ Spanning Tree Protocol (1)
      Protocol Identifier: Spanning Tree Protocol (0x0000)
      Protocol Version Identifier: Spanning Tree (0)
      BPDU Type: Configuration (0x00)
      BPDU flags: 0x00
      Root Identifier: 32768 / 10 / 00:1a:2b:3c:4d:01 (2)
      Root Path Cost: 0 (3)
      Bridge Identifier: 32768 / 10 / 00:1a:2b:3c:4d:01 (4)
      Port identifier: 0x8001 (5)
      Message Age: 0
      Max Age: 20
      Hello Time: 2 (6)
      Forward Delay: 15
      Originating VLAN (PVID): 10
```

#	What it denotes
1	Configuration BPDU. Only the root generates them; other switches relay them with updated cost.
2	**Who the sender thinks the root is.** All switches should advertise the same root ID. Format: priority / VLAN (sys-id-ext) / MAC.
3	**Cost to reach root** from the sender. Root sends 0; the neighbour (packet 2) sends 4 (a 1 Gbps link).
4	**Sender's own bridge ID.** Equal to root ID in packet 1 means this is the root bridge.
5	Port ID of the sending port: priority 128 (0x80) + port number.
6	Hello every 2 s. If BPDUs stop for max-age (20 s) the neighbour assumes the root is gone and recalculates. Packet 5: a switch with priority 4096 and the **TC flag** appears mid-capture and becomes root, so the topology changes (a classic root-hijack/ rogue switch or someone added a switch with lower priority). Mitigate with Root Guard / BPDU Guard.

Figure 7.2: PVST+ BPDUs for VLAN 10. Multiple 'TC' (Topology Change) BPDUs repeatedly means a flapping port is flushing MAC tables, causing unknown unicast flooding.

Wireshark: what a layer-2 loop / broadcast storm looks like

loop.pcapng - Wireshark

Apply a display filter ... `arp && eth.src==00:50:56:aa:11:22`

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000091	00:50:56:aa:11:22	Broadcast	ARP	64	Who has 192.168.10.77? Tell 192.168.10.10
2	0.000182	00:50:56:aa:11:22	Broadcast	ARP	64	Who has 192.168.10.77? Tell 192.168.10.10
3	0.000273	00:50:56:aa:11:22	Broadcast	ARP	64	Who has 192.168.10.77? Tell 192.168.10.10
4	0.000364	00:50:56:aa:11:22	Broadcast	ARP	64	Who has 192.168.10.77? Tell 192.168.10.10
5	0.000455	00:50:56:aa:11:22	Broadcast	ARP	64	Who has 192.168.10.77? Tell 192.168.10.10
6	0.000546	00:50:56:aa:11:22	Broadcast	ARP	64	Who has 192.168.10.77? Tell 192.168.10.10
7	0.000637	00:50:56:aa:11:22	Broadcast	ARP	64	Who has 192.168.10.77? Tell 192.168.10.10
8	0.000700	00:50:56:cc:00:01	Broadcast	ARP	64	Who has 192.168.10.77? Tell 192.168.10.10 [identical frame seen ag…

```
▸ Frame 3: 64 bytes on wire
▾ Ethernet II, Src: 00:50:56:aa:11:22, Dst: ff:ff:ff:ff:ff:ff (1)
▾ Address Resolution Protocol (request) (2)
      Sender MAC address: 00:50:56:aa:11:22
      Target IP address: 192.168.10.77
  [Frame time delta from previous displayed frame: 0.000091000 seconds] (3)
```

#	What it denotes
1	The **same source MAC and same frame** repeats endlessly. A single ARP request should appear once, not hundreds of times.
2	Identical content, tens of thousands of times per second (look at the time column: 91 µs apart). Check Statistics → I/O Graph, a flat-lined 100% line.
3	Time delta in microseconds. Real loops saturate links, CPU hits 100%, MAC addresses flap between ports.

Figure 7.3: Storm signature. On the switch you would also see: %SW_MATM-4-MACFLAP_NOTIF: Host 0050.56aa.1122 in vlan 10 is flapping between port Gi1/0/5 and port Gi1/0/6.

Loop troubleshooting steps	
1. Confirm	Interface utilization 100% on many ports, CPU high, MAC flap logs, users disconnected, STP topology change counter increasing (`show spanning-tree detail | inc ieee|occurr|from|is exec`).
2. Locate	`show interfaces | include line protocol|input rate|broadcasts` : ports with abnormal broadcast rates. Follow the MAC flap logs to the two ports.
3. Contain	Shut the offending port/uplink; unplug the looping cable (often an unmanaged mini-switch or patch-cable loop).
4. Prevent	BPDU Guard on access ports, storm-control, root bridge correctly set, no STP disabled, UDLD, document changes.

7.3 EtherChannel (LACP/PAgP)

- Bundles multiple physical links into one logical link: more bandwidth, redundancy, STP sees one link (no blocked members).
- **LACP (802.3ad, standard):** *active* / *passive*. **PAgP (Cisco):** *desirable* / *auto*. **Static:** *on* (no negotiation, risky). Active+active, active+passive, desirable+desirable or desirable+auto form a channel; passive+passive and auto+auto do not.
- Members must match: speed, duplex, VLAN/trunk mode, allowed VLANs, native VLAN, STP settings.
- Load balancing is per-flow hash (src/dst MAC, IP or L4 ports, `port-channel load-balance src-dst-ip`). One flow cannot exceed one member's speed.

```
interface range g1/0/47 - 48
 channel-group 1 mode active
interface port-channel 1
 switchport mode trunk
 switchport trunk allowed vlan 10,20

show etherchannel summary
Group  Port-channel  Protocol    Ports
1      Po1(SU)         LACP      Gi1/0/47(P)  Gi1/0/48(P)
! Flags: S = Layer2, U = in use, P = bundled, D = down, I = stand-alone, s = suspended, H = hot-standby
show etherchannel port-channel ; show lacp neighbor
```

EtherChannel problem	Cause / fix
Member shows (s) suspended or (I) stand-alone	Mismatch with partner (mode, speed, trunk/allowed VLANs) or partner not configured; compare `show run int` on both ends.
Po1 (SD)	Down: no active members; check physical links and LACP.
Uneven utilization	Hash algorithm; change load-balance method.
%PM-4-ERR_DISABLE: channel-misconfig	Both sides inconsistent (one 'on', the other LACP). Use same protocol.

7.4 Switch security essentials

```
! Port security
interface g1/0/5
 switchport port-security
 switchport port-security maximum 2
 switchport port-security mac-address sticky
 switchport port-security violation restrict      ! shutdown (default) | restrict | protect
! DHCP snooping + Dynamic ARP Inspection + IP Source Guard
ip dhcp snooping ; ip dhcp snooping vlan 10
ip arp inspection vlan 10
interface g1/0/48 ; ip dhcp snooping trust ; ip arp inspection trust
interface g1/0/5 ; ip verify source
show port-security interface g1/0/5 ; show ip arp inspection statistics
```

Attack	Mitigation
MAC flooding (CAM overflow)	Port security, max MAC
Rogue DHCP server	DHCP snooping (trusted ports only)
ARP spoofing	DAI (uses snooping bindings)
VLAN hopping (DTP switch spoofing, double tagging)	Disable DTP, set ports to access, unused VLAN as native, shut unused ports in a black-hole VLAN
Rogue switch / STP manipulation	BPDU Guard, Root Guard

7.5 Command cheat sheet

Goal	Command
Interface state, speed, duplex, VLAN	`show interfaces status`
Errors and drops	`show interfaces g1/0/1` (CRC, input errors, runts, giants, late collisions, output drops)
VLAN membership	`show vlan brief`
Trunk status / allowed VLANs	`show interfaces trunk`, `show interfaces g1/0/48 switchport`
MAC address table	`show mac address-table [dynamic] [vlan 10] [interface g1/0/5]`
Neighbours	`show cdp neighbors detail`, `show lldp neighbors`
Logs	`show logging | include %SW|%LINK|%SPANTREE|%PM`
SPAN / mirror for Wireshark	`monitor session 1 source interface g1/0/5 both` + `monitor session 1 destination interface g1/0/10`

7.6 Switching troubleshooting scenarios

Scenario	Method
User cannot reach anything; port is up/up	Verify VLAN of port (`show vlan brief`), VLAN exists, trunk carries it, gateway SVI up/up, port security, DHCP snooping, ARP to gateway in capture.
Port is err-disabled	`show interfaces status err-disabled` , find cause (bpduguard, port-security, link-flap, psecure-violation). Fix the cause, then shutdown/no shutdown.
Link up but slow, many CRC / late collisions	Speed/duplex mismatch (one side auto, other hard-coded), bad cable/SFP, EMI. Set both to auto, replace cable, check optical power (`show interfaces transceiver detail`).
VLAN 20 works on SW1 but not on SW2	VLAN missing in SW2's database, not in trunk allowed list, native mismatch, or VTP pruning/mode. Check each hop with `show interfaces trunk` .
CDP log: Native VLAN mismatch	Set the same native VLAN on both trunk ends.
Whole floor drops for ~30 s repeatedly	STP topology changes: flapping port, no PortFast on edge ports, or root bridge moving. Check TC counters and which port is flapping.
Trunk won't come up between two switches	Mode mismatch (access vs trunk, both dynamic auto), encapsulation, native VLAN, allowed VLAN, port-channel inconsistency.
Intermittent connectivity, MAC flapping logs	Loop, duplicate MAC, VM migration, or a hypervisor/NIC teaming without LACP. Identify the two ports in the log.
Packets drop at 1500+ bytes	MTU mismatch on trunk or jumbo mismatch; check `show system mtu` and path.

SVI up/down	No active port in VLAN or trunk not carrying it; VLAN shutdown; fix and verify `show ip interface brief`.

INTERVIEW ANGLE

"Explain how STP elects the root and decides which port blocks." Lowest bridge ID (priority + VLAN + MAC) wins root. Every other switch chooses one root port (lowest cost to root, then lowest sender bridge ID, then lowest port ID). On each segment one designated port is elected the same way; ports that are neither root nor designated block (alternate). Then say PortFast/BPDU guard on edge ports and Rapid-PVST for fast convergence.

8. Routing: static, OSPF, BGP, HSRP

How routers decide, and why adjacencies fail

8.1 Routing fundamentals

- **Route selection order:** (1) **longest prefix match** always wins first, (2) lowest **administrative distance** (trust of the source) among equal prefixes, (3) lowest **metric** within the same protocol. Equal metrics → ECMP load balancing.
- **Default route** 0.0.0.0/0 is the shortest possible match, used only when nothing more specific exists.
- **Floating static route:** static with higher AD (e.g. 200) as backup that appears only when the primary route disappears.

Source	AD	Source	AD
Connected	0	OSPF	110
Static	1	IS-IS	115
eBGP	20	RIP	120
EIGRP (internal)	90	iBGP	200

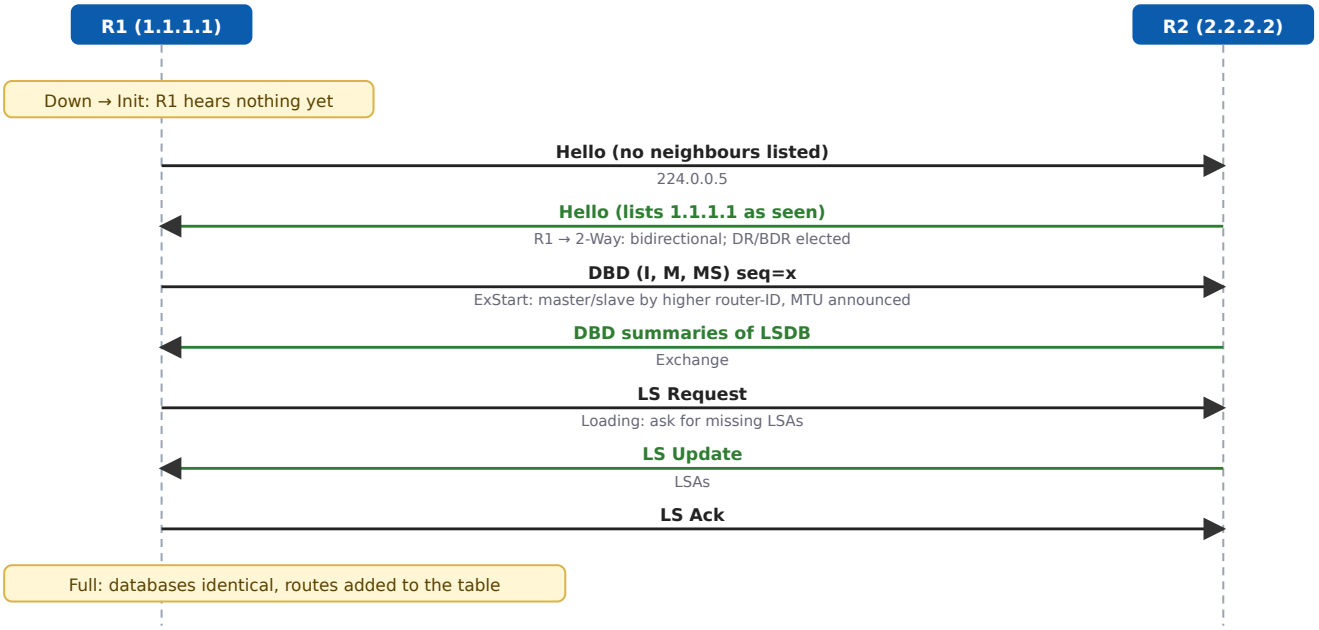
```
R1# show ip route
Gateway of last resort is 203.0.113.1 to network 0.0.0.0
S*    0.0.0.0/0 [1/0] via 203.0.113.1                <- static default, AD 1 metric 0
      10.0.0.0/8 is variably subnetted, 3 subnets
C        10.1.1.0/24 is directly connected, GigabitEthernet0/1
O        10.2.2.0/24 [110/20] via 10.1.1.2, 00:12:44, GigabitEthernet0/1   <- [AD/metric]
B     198.51.100.0/24 [20/0] via 203.0.113.2, 00:03:10
! add routes
ip route 10.50.0.0 255.255.0.0 10.1.1.2
ip route 10.50.0.0 255.255.0.0 172.31.0.2 200      ! floating backup
show ip route 10.2.2.5 ; show ip route summary ; show ip cef 10.2.2.5
```

8.2 OSPF

- **Link-state** protocol, IP protocol 89, multicast 224.0.0.5 (all OSPF routers) and 224.0.0.6 (DR/BDR). Metric = **cost** = reference bandwidth / interface bandwidth (default reference 100 Mbps: so Fast and Gigabit both cost 1; set `auto-cost reference-bandwidth 100000`).
- **Areas:** Area 0 (backbone) is mandatory; all other areas connect to it. ABR joins areas, ASBR redistributes external routes.
- **Router ID:** manual > highest loopback IP > highest active interface IP. Duplicate router IDs break adjacencies.
- **DR/BDR:** on broadcast networks, elected by highest priority (default 1, 0 = never) then highest router ID; not pre-emptive. Reduces adjacencies; point-to-point links have no DR.
- **Timers:** Hello 10 s / Dead 40 s on broadcast and point-to-point (30/120 on NBMA). Must match between neighbours.
- **LSAs:** Type 1 router, 2 network (from DR), 3 summary (between areas), 4 ASBR summary, 5 external, 7 NSSA external.

Neighbour state machine

OSPF adjacency states (Down, Init, 2-Way, ExStart, Exchange, Loading, Full)



Wireshark: OSPF Hello and adjacency formation

ospf.pcapng - Wireshark

Apply a display filter ... ospf

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	192.168.12.1	224.0.0.5	OSPF	78	Hello Packet
2	0.480000	192.168.12.2	224.0.0.5	OSPF	82	Hello Packet
3	1.100000	192.168.12.1	224.0.0.5	OSPF	82	Hello Packet
4	1.200000	192.168.12.2	192.168.12.1	OSPF	82	DB Description
5	1.201000	192.168.12.1	192.168.12.2	OSPF	82	DB Description
6	1.300000	192.168.12.1	192.168.12.2	OSPF	90	LS Request
7	1.301000	192.168.12.2	192.168.12.1	OSPF	146	LS Update
8	1.400000	192.168.12.1	224.0.0.5	OSPF	94	LS Acknowledge

```
▾ Internet Protocol Version 4, Src: 192.168.12.1, Dst: 224.0.0.5
      Time to Live: 1 (1)
      Protocol: OSPF IGP (89) (2)
      Differentiated Services Field: 0xc0 (DSCP: CS6)
▾ Open Shortest Path First
    ▾ OSPF Header (3)
          Version: 2
          Message Type: Hello Packet (1)
          Packet Length: 48
          Source OSPF Router: 1.1.1.1 (4)
          Area ID: 0.0.0.0 (Backbone) (5)
          Auth Type: Null (0)
    ▾ OSPF Hello Packet
          Network Mask: 255.255.255.0 (6)
          Hello Interval [sec]: 10 (7)
          Options: 0x12, (L) LLS Data block, (E) External Routing
          Router Priority: 1
          Router Dead Interval [sec]: 40 (7)
          Designated Router: 192.168.12.2
          Backup Designated Router: 0.0.0.0
          Active Neighbor: 2.2.2.2 (8)
```

#	What it denotes
1	TTL is 1: OSPF only talks to directly connected neighbours. A neighbour more than one hop away needs a virtual link or tunnel.
2	OSPF is its own IP protocol (89), not TCP/UDP, so port-based ACLs/firewalls do not see ports. Firewalls must allow protocol 89 and multicast.
3	The header: version 2 (OSPFv3 is for IPv6), message type 1-5 (Hello, DBD, LSR, LSU, LSAck).
4	**Router ID.** If two routers show the same ID, adjacency will not complete properly.
5	**Area ID** must match on both ends of the link.
6	**Network mask** must match on broadcast links (subnet mismatch = no adjacency).
7	**Hello/Dead timers** must match exactly. Mismatch = hellos are ignored; neighbours never form.
8	**Active Neighbor list.** If your router ID is listed in the neighbour's hello, it hears you (bidirectional). Seeing hellos from the neighbour that never list you = Init state: your hello is not reaching them (ACL, authentication, multicast filtering).

Figure 8.1: Hello (packet 3 selected) then DB Description, LS Request, LS Update, LS Acknowledge, which is the Exchange/Loading sequence before Full.

OSPF state stuck	Meaning / causes	Fix
No neighbour at all	Not receiving hellos or parameters mismatch: area ID, subnet/mask, hello/dead, authentication, stub flag, passive interface, ACL blocking 224.0.0.5, interface down, wrong network type	Compare `show ip ospf interface g0/0` on both sides; Wireshark the hello and compare fields
INIT	One-way: I hear them but they do not hear me	ACL/firewall in one direction, authentication one side, multicast filtering
2-WAY	Normal between DROther routers. Abnormal otherwise: both priority 0	Check priorities / DR election
EXSTART / EXCHANGE	**MTU mismatch** (most classic), duplicate router IDs, unicast blocked, DBD flapping	Match MTU or `ip ospf mtu-ignore`; unique router IDs
LOADING	Corrupt/missing LSAs, bad LS request	Clear process, check logs and link errors
FULL/DR or FULL/BDR	Healthy	—

```
router ospf 1
 router-id 1.1.1.1
 auto-cost reference-bandwidth 100000
 passive-interface default
 no passive-interface GigabitEthernet0/0
 network 192.168.12.0 0.0.0.255 area 0
 network 10.1.1.0 0.0.0.255 area 0
 default-information originate
interface g0/0
 ip ospf priority 100                    ! prefer as DR (needs re-election)
 ip ospf message-digest-key 1 md5 S3cr3t ! authentication (area/interface)

show ip ospf neighbor
Neighbor ID  Pri  State     Dead Time  Address       Interface
2.2.2.2        1  FULL/DR   00:00:35   192.168.12.2  Gi0/0
show ip ospf interface brief ; show ip ospf database ; show ip route ospf
debug ip ospf adj      ! use carefully in production, then undebug all
```

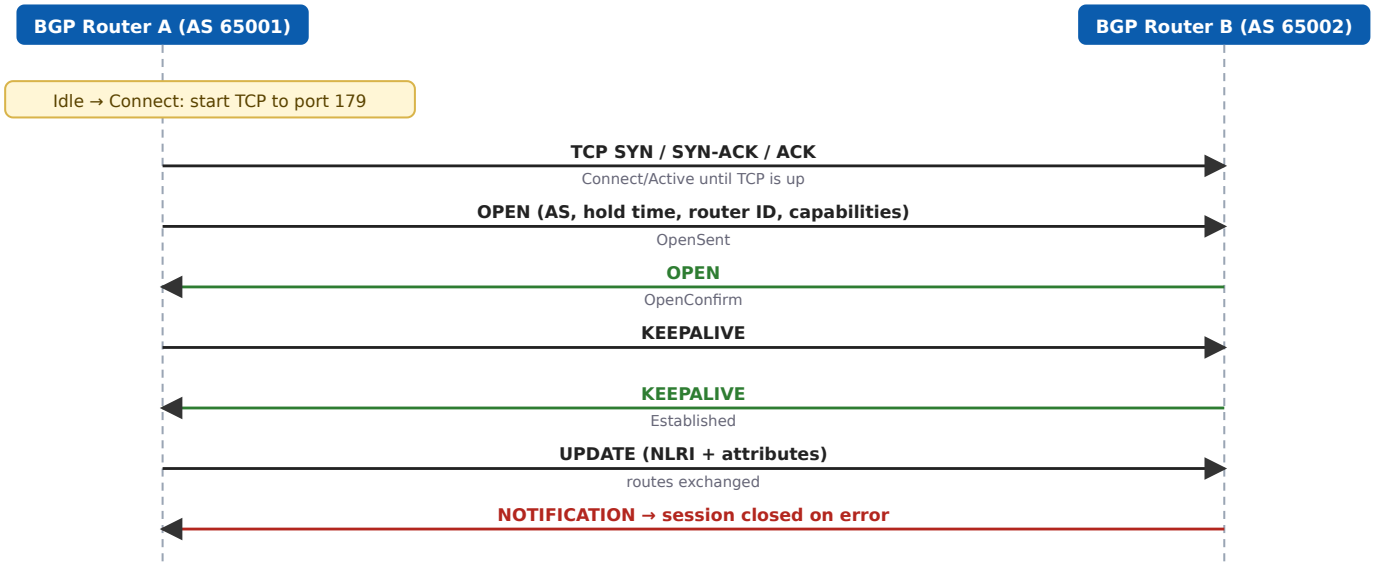
8.3 BGP

- **Path-vector** exterior routing protocol between autonomous systems (AS). Runs over **TCP 179** between explicitly configured neighbours (unicast, not multicast). Timers: keepalive 60 s, hold 180 s.
- **eBGP** between different AS (AD 20, TTL 1 by default so neighbours must be directly connected unless `ebgp-multihop`). **iBGP** within an AS (AD 200). iBGP does not re-advertise iBGP-learned routes (needs full mesh or route reflectors); typically use loopbacks with `update-source` and `next-hop-self`.
- **AS_PATH** also prevents loops: a router rejects routes containing its own AS.

#	Best-path criterion	Prefers
1	Weight (Cisco-local)	Highest
2	Local preference (shared in AS)	Highest
3	Locally originated	Yes
4	AS_PATH length	Shortest
5	Origin	IGP < EGP < Incomplete
6	MED (hint to neighbouring AS)	Lowest
7	eBGP over iBGP	eBGP
8	IGP metric to next hop	Lowest
9	Router ID	Lowest

Memory aid: Weight, Local-pref, Originated, AS-path, Origin, MED, eBGP, IGP metric, Router-ID. Outbound traffic: change local-pref. Inbound traffic: AS-path prepending or MED.

BGP FSM: Idle, Connect, Active, OpenSent, OpenConfirm, Established



Wireshark: BGP session, UPDATE and NOTIFICATION

bgp.pcapng - Wireshark

Apply a display filter ... `bgp || tcp.port==179`

No.	Time	Source	Destination	Protocol	Len	Info
1	0.000000	203.0.113.1	203.0.113.2	TCP	74	49876 → 179 [SYN] Seq=0 Win=16384 Len=0 MSS=1460
2	0.001500	203.0.113.2	203.0.113.1	TCP	74	179 → 49876 [SYN, ACK] Seq=0 Ack=1 Win=16384 Len=0 MSS=1460
3	0.001700	203.0.113.1	203.0.113.2	TCP	66	49876 → 179 [ACK] Seq=1 Ack=1 Win=16384 Len=0
4	0.002000	203.0.113.1	203.0.113.2	BGP	99	OPEN Message
5	0.003200	203.0.113.2	203.0.113.1	BGP	99	OPEN Message
6	0.003500	203.0.113.1	203.0.113.2	BGP	73	KEEPALIVE Message
7	0.003900	203.0.113.2	203.0.113.1	BGP	73	KEEPALIVE Message
8	0.250000	203.0.113.2	203.0.113.1	BGP	101	UPDATE Message
9	61.000000	203.0.113.2	203.0.113.1	BGP	75	NOTIFICATION Message (Error: Hold Timer Expired)

```
▸ Transmission Control Protocol, Src Port: 179, Dst Port: 49876
▾ Border Gateway Protocol - UPDATE Message (1)
      Marker: ffffffffffffffffffffffffffffffff
      Length: 47
      Type: UPDATE Message (2)
      Withdrawn Routes Length: 0
      Total Path Attribute Length: 20
    ▾ Path attributes (2)
      ▾ Path Attribute - ORIGIN: IGP
      ▾ Path Attribute - AS_PATH: 65002 (3)
            Type Code: AS_PATH (2)
            AS Path Segment: 65002
      ▾ Path Attribute - NEXT_HOP: 203.0.113.2 (4)
    ▾ Network Layer Reachability Information (NLRI) (5)
        198.51.100.0/24
```

#	What it denotes
1	Message type: OPEN(1), UPDATE(2), NOTIFICATION(3), KEEPALIVE(4). Everything runs inside one TCP session on port 179, so BGP problems often start as TCP problems (see chapter 4).
2	Path attributes describe how to reach the prefixes.
3	**AS_PATH**: list of ASes traversed. Used for loop prevention and best-path selection.
4	**NEXT_HOP** must be reachable, otherwise the route is not installed (common with iBGP).
5	**NLRI** = the prefixes being advertised in this update (198.51.100.0/24). A withdrawn route shows in Withdrawn Routes.

Figure 8.2: BGP session bring-up. In packet 9 the NOTIFICATION means the session is torn down (error code/subcode in details).

bgp-open.pcapng - Wireshark

Apply a display filter ... `bgp.type==1 || bgp.type==3`

No.	Time	Source	Destination	Protocol	Len	Info
4	0.002000	203.0.113.1	203.0.113.2	BGP	99	OPEN Message (My AS: 65001)
5	0.003200	203.0.113.2	203.0.113.1	BGP	99	OPEN Message (My AS: 65099 — but we expected 65002)
6	0.003600	203.0.113.1	203.0.113.2	BGP	75	NOTIFICATION Message OPEN Message Error / Bad Peer AS

```
▾ Border Gateway Protocol - OPEN Message
      Marker: ffffffffffffffffffffffffffffffff
      Length: 45
      Type: OPEN Message (1)
      Version: 4
      My AS: 65001 (1)
      Hold time: 180 (2)
      BGP identifier: 1.1.1.1 (3)
      Optional parameters length: 16
    ▾ Optional parameters
      ▾ Capabilities Advertisement (Length: 14) (4)
            Multiprotocol extensions capability (AFI: IPv4, SAFI: Unicast)
            Route refresh capability
            Support for 4-octet AS number capability (AS: 65001)
```

#	What it denotes
1	**My AS** must equal the AS the neighbour is configured with (`neighbor x remote-as`). A mismatch produces NOTIFICATION: Bad Peer AS, and the state falls back to Idle/Active.
2	Hold time: the lower of the two peers' values is used. Keepalive = hold/3.
3	BGP router ID, must be unique, otherwise Bad BGP Identifier.
4	Capabilities: both sides must agree on address families (AFI/SAFI). Missing MP-BGP IPv4 unicast or 4-byte AS support can prevent a session or prefix exchange.

Figure 8.3: OPEN mismatch. Packet 6 is the NOTIFICATION. Other notification codes: Message Header Error (1), OPEN Error (2), UPDATE Error (3), Hold Timer Expired (4), FSM Error (5), Cease (6).

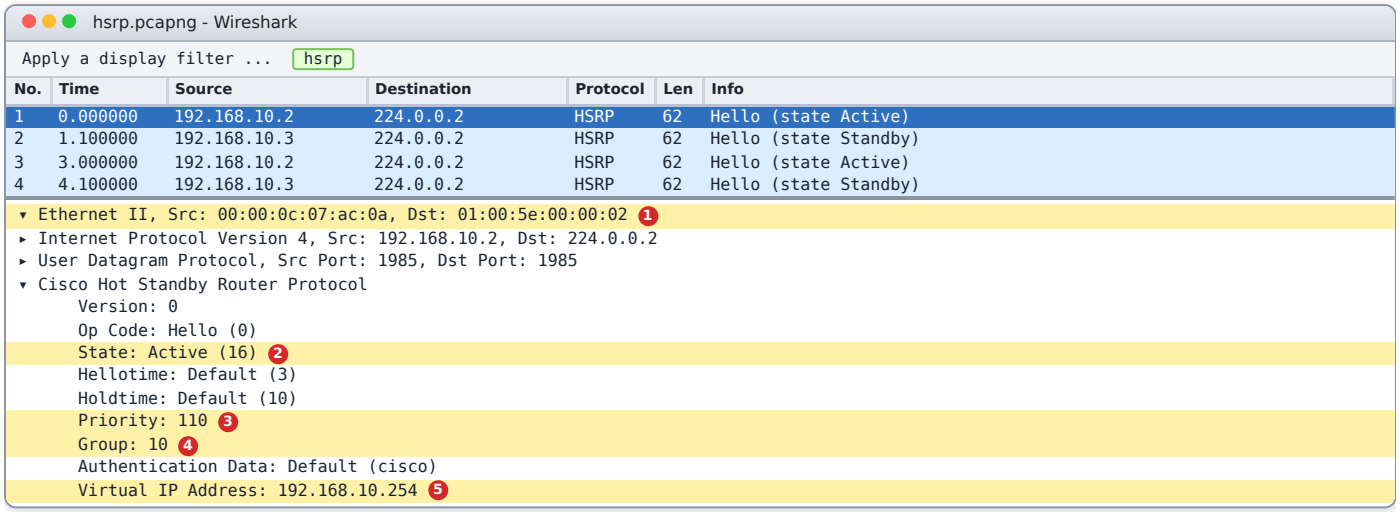
BGP state shown	Meaning	Check

Idle	Not trying (no route to neighbour, admin shutdown, previous fatal error, max-prefix hit)	Route to neighbour IP, `show ip bgp neighbors x.x.x.x` last error, shutdown, prefix limit
Active	Trying to establish TCP but failing (the name is misleading)	Reachability, ACL/firewall on TCP 179, wrong neighbour IP, other side not configured, update-source/ebgp-multihop, TTL
OpenSent / OpenConfirm	TCP up; OPEN problem	AS number mismatch, authentication (MD5) mismatch, capability mismatch, hold timers
Established (n)	Up; *n* = prefixes received	If 0: inbound filters, peer not advertising, network statement route missing from RIB, next-hop

```
router bgp 65001
 bgp router-id 1.1.1.1
 neighbor 203.0.113.2 remote-as 65002
 neighbor 203.0.113.2 description ISP-A
 neighbor 203.0.113.2 password S3cr3t
 neighbor 203.0.113.2 prefix-list FROM-ISP in
 network 198.51.100.0 mask 255.255.255.0        ! must exist in the routing table
show ip bgp summary
Neighbor     V   AS  MsgRcvd MsgSent TblVer InQ OutQ Up/Down  State/PfxRcd
203.0.113.2  4 65002     120     118      9   0    0 01:58:10 5
show ip bgp ; show ip bgp 198.51.100.0/24 ; show ip bgp neighbors 203.0.113.2 advertised-routes
clear ip bgp 203.0.113.2 soft in
```

8.4 HSRP / VRRP (first-hop redundancy)

- Two routers/L3 switches share a **virtual IP (VIP)** used as the hosts' default gateway. One is Active/Master (forwards), the other Standby/Backup. If Active fails, Standby takes the VIP and virtual MAC and sends a gratuitous ARP.
- **HSRP** (Cisco): priority default 100 (higher wins), tie → higher IP. **Preempt** must be configured for a recovered higher-priority router to take back the Active role. Hello 3 s, hold 10 s. v1 multicast 224.0.0.2, v2 224.0.0.102, UDP 1985. Virtual MAC v1 `0000.0c07.acXX` (XX = group in hex).
- **VRRP** (standard): IP protocol 112, multicast 224.0.0.18, virtual MAC `0000.5e00.01XX`. **GLBP** adds load sharing.
- Track an uplink so priority decreases if the WAN/uplink fails, to avoid black-holing traffic.



#	What it denotes
1	Active router sources hellos from the **virtual MAC** (…ac0a: group 10 = 0x0a), the Standby uses its own burned-in MAC.
2	State 16 = Active, 8 = Standby, 4 = Speak, 2 = Listen. If both routers claim Active, there is a **split-brain**: hellos not reaching each other (VLAN/trunk/STP problem).
3	Priority decides the Active router (higher wins).
4	Group number must match on both devices, and virtual MAC derives from it.
5	The **VIP** hosts use as the gateway. Must be in the subnet and match on both.

Figure 8.4: HSRP hellos every 3 s. If hellos stop for 10 s (hold time), Standby becomes Active.

```
interface Vlan10
 ip address 192.168.10.2 255.255.255.0
 standby 10 ip 192.168.10.254
 standby 10 priority 110
 standby 10 preempt
 standby 10 track 1 decrement 20          ! object tracking of uplink
show standby brief
```

8.5 Routing scenarios

Scenario	Troubleshooting
OSPF neighbour stuck in EXSTART	MTU mismatch on the link or SVI vs routed interface; compare `show interface` MTU; fix MTU or `ip ospf mtu-ignore`.
OSPF neighbours up but route missing	Network statement missing/wrong, passive interface, area mismatch for summary, filtering (distribute-list), route not originated; check LSDB with `show ip ospf database`.
BGP in Active	TCP/179 not completing: ping neighbour, check the correct source interface, ACLs/firewall, configuration on peer, ebgp-multihop for loopbacks.
BGP up but prefix not received	Peer isn't advertising (their network statement or filter), our inbound prefix-list/route-map, next-hop unreachable (iBGP), AS-path loop.
Asymmetric routing causing firewall drop	Request goes through FW1, reply returns via FW2 (no session). Fix routing or use symmetric return; PAN logs show flow_tcp_non_syn drops.
After HSRP failover users lose connection for a while	Switch MAC table not updated (no gratuitous ARP), STP convergence, preempt delay, uplink not tracked; check `show standby` logs.
Ping to destination works one way only	Return-route missing on the far side (no route back to source subnet) or NAT. Test with extended ping from the correct source interface.
Traffic taking suboptimal path	Check AD/metric/local-pref; use `show ip route x.x.x.x`, `traceroute`, check floating static is not active.

INTERVIEW ANGLE

"OSPF neighbour is not forming. What do you check?" Layer 1/2 reachability and IPs on the same subnet, then hello parameters (area, timers, mask, authentication, stub flags, network type), passive-interface, ACL/firewall allowing protocol 89 and multicast, MTU if stuck in ExStart, duplicate router ID. Verify with *show ip ospf interface* and a hello capture.